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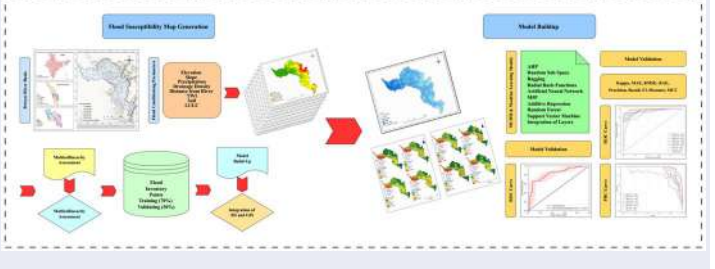
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ABSTRACT

Floods have become a significant global concern in recent times due to their devastating impact to infrastructure and society. The key flood-conditioning factors such as rainfall, elevation, slope, drainage density, distance from the river, TWI, soil type, and LULC were assigned weightage using the AHP method. Several machine learning models were employed to examine the relationship between flood occurrences and susceptibility. The flood risk zones were classified into five categories: very high (4%), high (14%), moderate (52%), low (16%), and very low (14%). The AUC-ROC results demonstrated high accuracy across different models: AHP (0.825), Random Subspace (0.974), Bagging (0.985), Radial Basis Function (0.937), Artificial Neural Network (0.953), M5P (0.946), Additive Regression (0.960), Random Forest (0.932), and Support Vector Machine (0.882). The findings of this study will assist relevant authorities in formulating flood mitigation strategies and supporting Sustainable Development Goals (SDGs).

GRAPHICAL ABSTRACT



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KEYWORDS
Flood; remote sensing; multi-criteria analysis; machine learning; AHP

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1. Introduction

Floods are among the most frequent and devastating natural disasters worldwide, causing immense damage to infrastructure, ecosystems, and human life (Aydin and Iban 2023). Floods have several positive effects on the environment, although at smaller scales. These include providing water and nutrients to riparian corridors, removing contaminants from flood plain areas, replenishing the groundwater table, and improving soil fertility (Johansen et al. 2015). Exceeding the flood tolerance limit will result in significant repercussions for human life, agricultural practices, residential areas, properties, roadways, and natural ecosystems (Dutta and Deka 2024). A stream or water body is usually considered flooded when its flow exceeds the capacity of the river banks and fills the nearby low-land regions, causing significant socioeconomic and environmental impacts. Urban expansion, uncontrolled, and natural factors like rainfall, deforestation, land-use patterns, and silt accumulation have increased vulnerability to flood damage and casualties in regions (Debnath et al. 2024). Flood is considered one of the most devastating disasters due to its recurrence, and its causes must be identified (Ke et al. 2020).

Accurate flood susceptibility mapping (FSM) is crucial for effective flood risk management, mitigation, and disaster preparedness (Tehrany et al. 2014; Rubinato et al. 2019; Rodriguez et al. 2021; Rosenzweig et al. 2021). Floods can be significantly reduced *via* careful analysis of their spatially characteristics. Dams and non-structural interventions, such as public awareness campaigns, legislative instruments, forecasts, early notice systems, and rescue support, can mitigate the adverse effects of flood disasters on people and assets (Büchele et al. 2006; Campbell et al. 2016; Mangukiya and Sharma 2022; Nsangou et al. 2022). GIS and remote sensing techniques are crucial for visualizing and mitigating flood-prone regions (Safaripour et al. 2012; Rahmati et al. 2016; Samanta et al. 2018; Siahkamari et al. 2018; Vaishnavi et al. 2020). They immediately help the decision-making process by enabling real-time forecasting, risk management, and management of land use and land cover (Khoirunisa et al. 2021; Ahire et al. 2022). Many researchers from all around the globe have utilized a wide variety of techniques in order to map flood hazards, including; Multi-Criteria Decision-Making (MCDM) (Asiri et al. 2024; Zewdu et al. 2024), Analytic Hierarchy Process (AHP) (Islam et al. 2021; Mitra et al. 2022; Senan et al. 2023; Aloui et al. 2024; Dutta and Deka 2024), Analytical Network Process (ANP) (Solaimani et al. 2023; Shah and Pan 2024), Complex Proportional Assessment (COPRAS) (Gholami et al. 2025; Tavosi et al. 2025), Criteria Importance Through Inter-criteria Correlation (CRITIC) (Rai et al. 2024), Decision-Making Trial and Evaluation Laboratory (DEMATEL) (Efraimidou and Spiliotis 2024; Jain et al. 2024), Evaluation Based on Distance from Average Solution (EDAS) (Shah and Pan 2024; Sarkar and Gayen 2025), Fuzzy Analytic Hierarchy Process (F-AHP) (Solaimani et al. 2023; Kumar and Jha 2024), Preference Ranking Organization METHod for Enrichment Evaluation (PROMETHEE) (Plataridis and Mallios 2024), Simple Additive Weighting (SAW) (Abassi et al. 2024; Tavosi et al. 2025), Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) (Mahmoodi et al. 2024; Plataridis and Mallios 2024), VlseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR) (Hamidifar et al. 2024; Jain et al. 2024), Weighted Overlay Analysis (WOA) (Koralay and Kara

2024; Mondal et al. 2024), Weighting Factor (WF) (Saha et al. 2024). Bivariate statistical models including; Certainty Factor (CF) (Chowdhury 2024), Evidential Belief Function (EBF) (Eghbali et al. 2024; Mandal et al. 2025), Frequency Ratio (FR) (Elghouat et al. 2024; Ghobadi and Ahmadipari 2024), Index of Entropy (IoE) (Costache et al. 2024; Osman and Das 2024), Shannon Entropy (Chetia and Paul 2024; Sharma et al. 2024), Statistical Index (SI) (Diaconu et al. 2024; Pawar 2024), Weights of Evidence (WoE) (Büchele et al. 2006; Islam et al. 2021; Behera et al. 2023; Mansour et al. 2024; Muthu and Ramamoorthy 2024), and Fuzzy Logic (Nguyen 2022).

Around the world, many techniques have been used to determine risk areas and flood threats including analytical hierarchy process (AHP) (Das 2018; Khan et al. 2025), Artificial Neural Network (ANN) (Islam et al. 2021; Ha et al. 2022), hydrological and hydrodynamic models (Islam et al. 2021; Abassi et al. 2024; Kumar et al. 2024), Multi-Criteria Decision Analysis (MCDA) (Gigović et al. 2017) and machine learning approaches (Feloni et al. 2020). Multi-criteria decision analysis (MCDA) is widely used method for flood-hazard zonation (Abdelkarim, et al. 2020, Vignesh, et al. 2021). The Analytic Hierarchy Process (AHP), developed by Saaty in 1980, is the most widely used approach among multi-criteria decision-making methodologies (MCDM). The Analytic Hierarchy Process (AHP) is a versatile mathematical model that takes into account the priorities of individuals or groups and assesses both quantitative and qualitative factors in decision-making situations (Bansal et al. 2022). The AHP environment analyzes multi-criteria layers to create a hierarchical structure that provides rankings and weights, with input from users and experts (Hussain et al. 2021; 2023). The seamless incorporation of Multiple Criteria Decision Analysis (MCDA) into Geographic Information Systems (GIS), along with the uniformity in evaluation, significantly enhances the decision-making process and has demonstrated considerable success in several research studies (Ogden et al. 2011; Patel et al. 2015; Costache 2019; Nigusse and Adhanom 2019; Souissi et al. 2020; Luu et al. 2021; Aydin and Iban 2023; Dahri et al. 2022; Gharakhanlou and Perez 2023; Moradian et al. 2024).

This study aims to apply the Analytic Hierarchy Process (AHP), Machine Learning (ML), Remote Sensing (RS), and Geographic Information System (GIS) techniques to assess the flood susceptibility and risk in the Periyar river basin, located in Kerala, India. It is the longest perineal river in Kerala, with a considerable discharge potential, principal source of drinking water, and energy production site. There has been a dramatic change in the amount of rainfall received and the frequency of flooding in Kerala since 2015. The catastrophic floods in 2018 and 2019 proved that low-lying, densely populated areas were the hardest hit. In view of the increased frequency of floods in the Periyar river basin, it is imperative to conduct a comprehensive flood study in order to identify the areas that are most susceptible to flooding. We have utilized eight criteria viz., rainfall, slope, elevation, LULC, distance from river, topographic wetness index, drainage density, and soil texture pertaining to elevation and other terrain-related characteristics for preparing the flood susceptibility map of the region.

With the advancement of geospatial technologies and computational models, Machine Learning (ML) has emerged as a powerful tool for predicting flood-prone

areas. The integration of RS and GIS with ML techniques enhances the accuracy and efficiency of flood susceptibility assessments (Al-Aizari et al. 2022; Ullah et al. 2022; Hussain et al. 2023; Al-Aizari et al. 2024; Rahman et al. 2024; Hussain et al. 2025). Flood susceptibility maps of the area are generated with eight sophisticated machine learning algorithms; Random Sub Space (Luu et al. 2021; Li and Hong 2023; Riazi et al. 2023; Dutta and Deka 2024), Bagging (Pham et al. 2021; Ha et al. 2022; Pal and Singha 2022; Shen et al. 2023), Radial Basis Functions (Pham et al. 2021; Nguyen 2022; Luu et al. 2023; Moradian et al. 2024), Artificial Neural Network (Khoirunisa et al. 2021; Dahri et al. 2022; Rudra and Sarkar 2023; Sharma et al. 2024), M5P (Talukdar et al. 2020; Rahman et al. 2021; Yukseler et al. 2023; Rostami et al. 2025), Additive Regression (Pal and Singha 2022), Random Forest (Aydin and Iban 2023; Sarkar et al. 2022; Gharakhanlou and Perez 2023; Debnath et al. 2024), Support Vector Machine (Saravanan and Abijith 2022; Mehravar et al. 2023; Halder et al. 2024; Hassan et al. 2025).

The present study also utilized Google Earth Engine (GEE) to map the flood event of 2018 using data from the Sentinel 1 Synthetic Aperture Radar (SAR) images. Google Earth Engine (GEE) is a cutting-edge geospatial platform that enhances satellite image processing by providing free satellite data, over 40 years of Earth's surface observations, and continuous updates on radar data quality and quantity (Otsu 1979; Helmrich et al. 2021; Debnath et al. 2023; Demissie et al. 2024). The pre-flood and post-flood scenarios of the 2018 flood were retrieved using the Google Earth Engine platform. The algorithm developed by Hassan et al. (2025) was utilized to determine the threshold for the flood pixel in the region under study. This allows us to examine how much water has been dispersed to metropolitan regions and how it has affected human life. Mapping flood-prone regions will aid urban planners, governments, and communities in establishing mitigation methods, and assisting local, national, and international organizations in reducing flood disaster risk. Flood risk management (FRM) is essential for sustainable development as floods can cause extensive damage to lives, property, infrastructure and ecosystems.

Although numerous studies have applied either Analytical Hierarchy Process (AHP) or individual Machine Learning (ML) algorithms for flood susceptibility mapping, limited research exists that integrates expert-based AHP with a comparative ensemble of multiple advanced ML models within a unified geospatial framework. Moreover, few studies have comprehensively evaluated flood vulnerability using a robust combination of eight critical flood conditioning parameters—rainfall, elevation, slope, drainage density, distance from river, topographic wetness index (TWI), soil type, and land use/land cover (LULC)—specifically for flood-prone and ecologically sensitive basins like the Periyar River Basin in Kerala, which face recurrent and intensifying flood events. This study addresses this gap by combining AHP with eight machine learning models (Random Sub Space, Bagging, Radial Basis Function, Artificial Neural Network, M5P, Additive Regression, Random Forest, and Support Vector Machine), integrated with GIS and RS techniques, to provide a novel, data-informed, and expert-driven approach for high-precision flood susceptibility mapping. Moreover, few studies utilize robust flood conditioning parameters in combination with cloud-based geospatial platforms like Google Earth Engine (GEE) for pre and

post-flood analysis using high-resolution SAR imagery. This hybrid methodology enhances both the reliability and practical applicability of flood risk assessment for sustainable disaster management and planning.

Several Sustainable Development Goals (SDGs) outlined by the United Nations directly or indirectly support flood risk reduction and management efforts. Flood susceptibility mapping plays a crucial role in achieving several targets of the United Nations Sustainable Development Goals (SDGs), particularly SDG 6 (Clean Water and Sanitation), SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), SDG 13 (Climate Action), SDG 14 (Life Below Water), and SDG 15 (Life on Land). SDG 6 promotes resilient water infrastructure and effective management of water-related ecosystems, which can be supported by identifying flood-prone zones for improved drainage and water resource planning. SDG 9 emphasizes the development of reliable infrastructure and the use of innovative technologies—such as integration of RS and GIS with knowledge based AHP and machine learning algorithms—for disaster-resilient development. SDG 11 advocates for inclusive, safe, and resilient human settlements; flood risk maps are essential tools for urban planning, zoning, and community safety. SDG 13 calls for urgent action to combat climate change and its impacts; mapping flood susceptibility enables climate adaptation by identifying high-risk areas vulnerable to extreme rainfall and hydrological events. SDG 14 and SDG 15, focusing on the conservation of aquatic and terrestrial ecosystems, respectively, are supported by flood mapping that informs sustainable watershed management, helps prevent habitat degradation, and guides land-use decisions to reduce environmental vulnerability. Thus, the integration of geospatial tools, AHP, and machine learning algorithms in this study not only enhances flood risk assessment in the Periyar River Basin but also contributes directly to sustainable development and climate resilience at the regional scale (Binns 2022; Elsharqawy et al. 2022; Dolatshahi and Kerachian 2023; Oneto and Canepa 2023; Chaudhary et al. 2024; Janizadeh et al. 2024; Leeonis et al. 2024; Nguyen et al. 2024; Wu et al. 2024; Yilmaz 2025; Zhou et al. 2024; Abijith et al. 2025; Akhter et al. 2025; Ibrahim et al. 2025; Joe et al. 2025; Mhangara et al. 2025; Rahman et al. 2025). Flood risk management aligns with multiple SDGs by promoting resilience, sustainability, and disaster preparedness. These approaches enable policymakers and stakeholders to make informed decisions, reducing flood-related vulnerabilities ensuring long-term long-term environmental and economic sustainability.

2. Study area

The Periyar river is one of the longest rivers in Kerala state lies between longitudes of 90°16′45″N and latitudes of 100°26′45″N and 76°07′28″E and 77°016′30″E. It begins from the Sivangiri peaks of 'Sundarmala', flows through the centre of Kerala, and empties into the Arabian Sea near Cochin. The whole length of the river is 300 km, of which 244 km flowing through Kerala and the rest through the neighbouring state of Tamil Nadu. The overall river catchment area is 6350 sq. km, while Kerala's is 5398 sq. km. The centre portion of the river is located in the districts of Idukki and Ernakulam, with a tiny portion passing through steep slopes and gorges

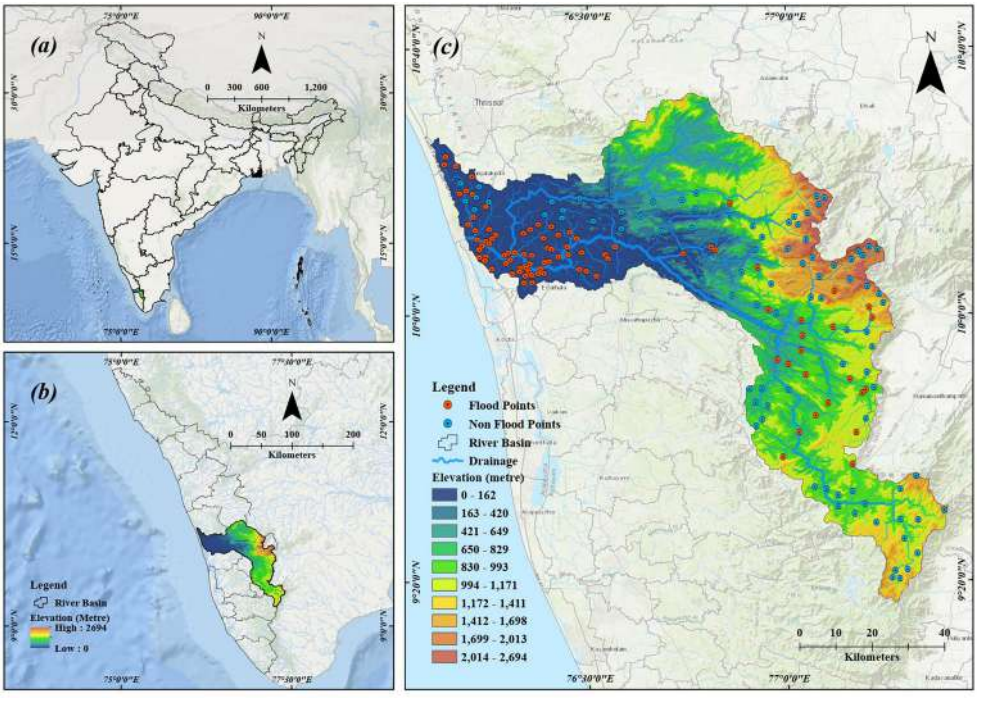


Figure 1. Location map of study area: (a) India, (b) Kerala, and (c) Periyar river basin.

in the district of Thrissur. Based on Figure 1, the basin has an inverted 'L' form and a dendritic pattern with maximal width in the centre. The upper stream of the Periyar river basin in the district of Idukki receives an average annual rainfall of 3,677 mm, while the lower stream in the district of Ernakulam receives 3,099 mm. This region's highest temperature ranges from 24 to 32.10 °C, while the minimum temperature is from 23 to 26 °C (Figure 2). The river basin features seventeen dams, the greatest of which is the Idukki reservoir, which has three dams, Idukki, Cheruthonni, and Idamalayar, for managing water in the reservoir and producing hydroelectricity. Figure 1 depicts the Periyar river basin and its drainage network.

3. Materials and methods

Identifying flood-triggering factors is critical to generating flood hazard maps and depends on the study area's geographical characteristics and data availability (Debnath et al. 2023). Rainfall, slope, elevation, LULC, distance from river, topographic wetness index, drainage density, and soil texture are the eight criteria that have been chosen for this study. These factors were chosen after conducting a comprehensive literature analysis (Ogden et al. 2011; Gharakhanlou and Perez 2023; Moradian et al. 2024) and mostly owing to data availability. Thematic layers were generated by integrating meteorological data, conventional maps, and remote sensing imagery from various organizations and portals. The Digital Elevation Model (DEM) obtained by NASA's Shuttle Radar Topographic Mission (SRTM) has a resolution of 30 meters and was acquired from the USGS Earth Explorer portal (https://earthexplorer.usgs.gov).

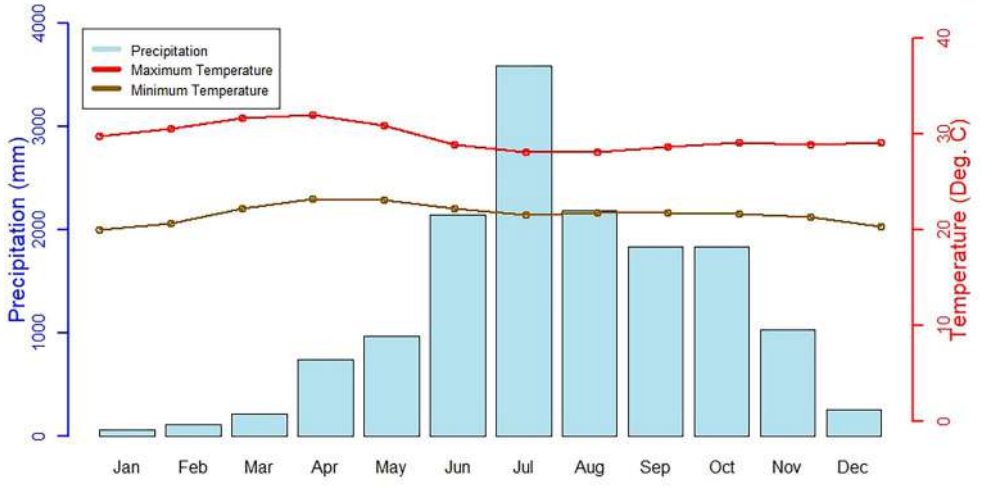


Figure 2. Climatic chart showing the rainfall, maximum and minimum temperature.

Table 1. Sources of data.

Data	Description	Scale or resolution	Source
SRTM Dem (2014)	SRTM1N09E076V3 SRTM1N09E077V3 SRTM1N10E076V3 SRTM1N10E077V3	1-ARC	https://earthexplorer.usgs.gov/
LULC (2021)	Esri Land cover-Sentinel 2	10 m	https://livingatlas.arcgis.com/landcover/
Soil (2003)	FAO World soil shapefile	1:5000000	https://www.fao.org/soils-portal/data-hub/soil-maps-and-databases/faounesco-soil-map-of-the-world/en/
Rainfall (2015, 2018,2021)	GPM (IMERG)	Spatial resolution-0.1	https://giovanni.gsfc.nasa.gov/giovanni/

The drainage density, slope, topographic wetness index (TWI) and elevation maps were generated in ArcGIS Pro using the SRTM DEM. The line density tool in ArcGIS is utilized in the process of generating the drainage density map. The slope is calculated from the Digital Elevation Model (DEM) data by utilizing the spatial analyst tool in ArcGIS. The rainfall map is prepared by utilizing IMERG, a NASA Level 3 GPM-Era global precipitation product, accessed from the Giovanni portal (https://disc.gsfc.nasa.gov/datasets/GPM_3IMERGM_06/summary?keywords=%22IMERG%20final%22). We prepared the soil map for the research area using the Food and Agriculture Organization's (FAO) World Soil Data (Table 1). Using the Sentinel 2 FCC satellite image from the study area, the land use/land cover map was generated using the random forest-supervised classification technique. The schematic diagram provides a detailed explanation of the approach used in the present research (Figure 3).

3.1. Multi-criteria decision analysis and analytical hierarchy process (AHP) method

The analytical hierarchy process (AHP) is a flexible and well-organized method for analyzing intricate decision scenarios that involve multiple aspects (Saaty 1987; Li and

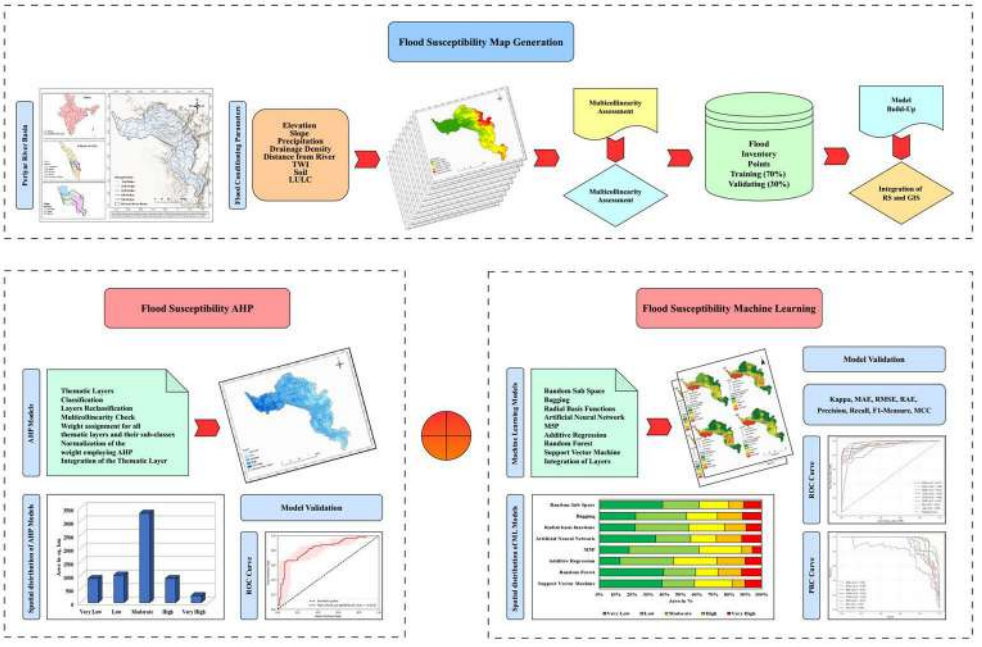


Figure 3. Methodology of the study.

Table 2. Saaty's nine-point scale for analytical Hierarchy Process (AHP) preference (Saaty 1980).

Rank	Scale
1	Equally important
2	Very strong to extremely
3	Moderately important
4	Strongly to very strongly
5	Strongly important
6	Moderately to strongly
7	Very Strongly important
8	Equally to moderately
9	Extremely preferred

Hong 2023). The MCDA technique is widely utilized for flood susceptibility mapping due to its global significance and relevance in current matters. The Multi-Criteria Decision Analysis approach, namely the Analytic Hierarchy Process (AHP), was used to allocate the rank for categories in a layer and the weights for individual components through pair-wise comparison matrix (Saaty 1980; Stefanidis and Stathis 2013). The flood susceptibility map of the region was generated by standardizing the weights and ratings assigned to the various flood conditioning factors. Each criterion is assigned a numerical value on a scale of 1 to 9, based on its importance, in square matrices (Table 2). Subsequently, the AHP method proposed by Saaty (1980) is employed to build pairwise comparison matrices for each thematic layer and subclass (Ahire et al. 2022; Behera et al. 2023). The weights are then normalized using the eigenvector technique. In order to evaluate the relative importance of various thematic layers and their specific categories, the consistency ratio (CR) was calculated based on the method proposed by Saaty (1987, 1980) and Li and Hong (2023).

3.2. Calculation of weightage using the AHP method

The Analytical Hierarchy Process (AHP) was developed by T. LSaaty in the late 1970s and is currently the most widely used model for ranking choice alternatives in the field of Multi Criteria Decision Analysis (MCDA) (Li and Hong 2023). It is an efficient method for solving complicated problems, and it is typically applied by ranking or weighting the components and their categories (Mokhtari et al. 2023). Because of its significance and applicability to current problems, the MCDA technique is widely employed in the mapping of flood vulnerability. Therefore, the AHP approach has been used in this study to integrate the chosen theme layers of flood vulnerability. Each layer's weight is proportional to the impact it has on the vulnerability mapping preparation. Using pair-wise comparison matrices, AHP compares the relative weight of each class within the same thematic layer as well as between thematic layers (Saaty 1980). Thus, the relative importance of each layer has been established by utilizing Saaty's preference scale, which ranges from 1 to 9 (Table 3), in conjunction with literature review, field knowledge, and studies conducted in comparable geographical regions.

In order to evaluate the pair-wise comparison of each parameter and its subcategories, the consistency ratio (CR) was applied. The CR should be less than 10% in order for matrix comparisons to be consistent and the AHP results to be acceptable. According to the aforementioned computation, the CR is 4.4, or 4%, which is less than 10%. The following equation is used to determine the

$$CR = CI/RI \tag{1}$$

$$CI = (max - n)/(n - 1) \tag{2}$$

where max is the maximum eigenvalues, n is the number of key variables, RI is the random inconsistency index, CR is the consistency ratio, and CI is the consistency index (Saaty and Vargas 1993).

3.3. Machine learning models

3.3.1. Random subspace (RSS)

Random Subspace (RSS) is an ensemble learning method that enhances flood susceptibility prediction by training multiple models on randomly selected input features

Table 3. Confusion matrix.

Criteria	TWI	Elevation	Slope	Precipitation	Distance from river	LULU	Drainage density	Soil
TWI	1	3	3	1	1	5	1	3
Elevation	1/3	1	3	1	1	3	1/3	3
Slope	1/3	1/3	1	1	1/5	5	1/3	3
Precipitation	1	1	1	1	1	5	1	3
Distance from River	1	1	5	1	1	7	1	5
LULC	1/5	1/3	1/3	1/5	1/7	1	1/7	1
Drainage Density	1	3	3	1	1	7	1	7
Soil	1/3	1/3	1/3	1/3	1/5	1	1/7	1

(Luu et al. 2021). It reduces overfitting, making it suitable for complex flood-prone regions (Li and Hong 2023; Riazi, et al. 2023). RSS can be combined with classifiers for improved performance and is effective in handling imbalanced datasets (Dutta and Deka 2024). Given dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$ with M features, RSS randomly selects m features $(m > d)$ and trains a base classifier $h_t(x)$.

$$h_t(x) = f(x_{t_1}, x_{t_2}, \cdots, x_{t_n}) \tag{3}$$

where x_{t_j} are the randomly selected features. The final flood susceptibility prediction is obtained by aggregating the outputs of T base models using,

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} h_t(x) \tag{4}$$

where $h_t(x)$ is the classifier trained on a subset of features, and T is the number of base classifiers. For regression, or majority voting for classification,

$$\hat{y} = \text{mode}(h_1(x), h_2(x), ..., h_T(x)) \tag{5}$$

where $h_t(x)$ is the model's prediction for each sample.

3.3.2. Bagging

Bagging (Bootstrap Aggregating) is an ensemble learning technique that enhances flood susceptibility predictions by combining multiple models trained on different data subsets (Pham et al. 2021). It generates multiple datasets, reducing variance and overfitting, especially in high-variance models like Decision Trees (Ha et al. 2022; Pal and Singha 2022). Bagging improves accuracy and robustness, especially in complex flood-prone areas. It is computationally efficient, efficient, and works well with imbalanced datasets (Shen et al. 2023). Given dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$ Bagging generates T bootstrap samples Dt from D and trains a base model $h_t(x)$ on each sample. The final prediction is given by, For regression,

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} h_t(x) \tag{6}$$

For classification,

$$\hat{y} = \text{mode}(h_1(x), h_2(x), ..., h_T(x)) \tag{7}$$

where $h_t(x)$ is the model's prediction for each sample.

3.3.3. Radial basis function (RBF)

The Radial Basis Function (RBF) is a machine learning model used for flood susceptibility mapping, capturing complex spatial patterns (Pham et al. 2021). It is sensitive to local variations in flood-prone areas and is highly adaptive, ensuring smooth

interpolation of flood hazard levels (Nguyen 2022; Luu et al. 2023). Hyperparameter tuning, such as adjusting the spread parameter, is crucial for improving model accuracy (Moradian et al. 2024). The Radial Basis Function is defined as,

$$\phi(x) = \exp\left(-\frac{\|x - c_i\|^2}{2\sigma^2}\right) \tag{8}$$

where x is the input feature vector, c is the centre of the RBF neuron, σ is the spread parameter, controlling the function's sensitivity, $\|x - c_i\|$ is the Euclidean distance between the input and the centre. The final flood susceptibility prediction is given by:

$$y(x) = \sum_{i=1}^{n} w\phi(x_i) + b \tag{9}$$

where w_i are the weights, and b is the bias term.

3.3.4. Artificial neural networks (ANN)

Artificial Neural Networks (ANN) are machine learning models that mimic the human brain's structure for flood susceptibility prediction (Khoirunisa et al. 2021). They consist of an input, hidden, and output layer, processing information using weighted connections (Dahri et al. 2022; Rudra and Sarkar 2023). ANN is robust against noisy or missing data, suitable for flood-prone areas, and requires hyperparameter tuning for optimal performance (Sharma et al. 2024). The output of an ANN neuron is given by,

$$y = f\left(\sum_{i=1}^{n} w_i x_i + b\right) \tag{10}$$

where x_i are input flood-related features, w_i are weights assigned to each feature, b is the bias term, $f(x)$ is the activation function, such as, Sigmoid function for classification,

$$f(x) = \frac{1}{1 - e^{-x}} \tag{11}$$

or, ReLU function for regression,

$$f(x) = \max(0, x) \tag{12}$$

This allows ANN to model flood susceptibility efficiently.

3.3.5. M5P

M5P is a hybrid machine learning model that uses decision trees and linear regression models to improve flood susceptibility prediction (Talukdar et al. 2020). It divides study areas into flood-prone and non-flood-prone regions, estimating numerical flood susceptibility using linear regression models (Rahman et al. 2021; Elsharqawy et al. 2022). M5P reduces overfitting and provides interpretable linear

equations for flood-prone areas, adapting to diverse environmental conditions (Rostami et al. 2025). M5P follows two main steps; (a) Decision Tree Splitting: The dataset is recursively partitioned based on flood-related attributes X, using variance reduction,

$$\Delta S = S_{parent} - \sum_{j} \frac{|S_j|}{|S_{parent}|} S_j \tag{13}$$

where S is the variance of flood susceptibility values. (b) Linear Regression at Leaf Nodes: Once a terminal node is reached, a linear regression model is fitted,

$$y = a_0 + a_1 x_1 + a_2 x_2 + \cdots + a_n x_n \tag{14}$$

where y is the flood susceptibility score, and a_i are regression coefficients for input variables x_i.

3.3.6. Additive regression (AR)

Additive Regression is an ensemble learning method that enhances flood susceptibility predictions by refining errors from previous models (Janizadeh et al. 2021). It focuses on difficult-to-predict flood-prone areas and reduces bias and variance risk. It performs well with spatially complex flood data and can be combined with decision trees or neural networks for enhanced flood prediction (Pal and Singha 2022). Given dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$, Additive Regression models the target variable as,

$$f(x) = f_0(x) + \sum_{m=1}^{M} \lambda_m h_m(x) \tag{15}$$

where $f_0(x)$ is the initial model, $h_m(x)$ is the weak regressor at iteration m, λ_m is the learning rate, M is the total number of weak regressors. Each new model $h_m(x)$ is trained to minimize the residual error of the previous step,

$$r_m = y - f_{m-1}(x) \tag{16}$$

where r_m represents the residual error.

3.3.7. Random forest (RF)

Random Forest (RF) is an ensemble learning algorithm that uses multiple decision trees to improve flood susceptibility prediction (Aydin and Iban 2023). It uses feature bagging, robust against overfitting, and can handle high-dimensional datasets (Sarkar et al. 2022; Gharakhanlou and Perez 2023). RF is suitable for complex, heterogeneous flood-prone regions and is computationally efficient due to its parallel processing capability (Debnath et al. 2024). Given dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$, RF constructs multiple decision trees $h_t(x)$ and the final prediction is obtained by:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} h_t(x) \tag{17}$$

where $h_t(x)$ is the prediction of the t-th decision tree, T is the total number of trees. For classification, majority voting is used. For regression, predictions are averaged.

$$\hat{y} = \text{mode}(h_1(x), h_2(x), ..., h_T(x)) \tag{18}$$

where $h_t(x)$ is the model's prediction for each sample.

3.3.8. Support vector machine (SVM)

Support Vector Machine (SVM) is a powerful supervised learning algorithm used for flood susceptibility mapping, classifying flood-prone and non-flood-prone areas (Saravanan and Abijith 2022). It handles high-dimensional flood datasets and incorporates parameters (Mehravar et al. 2023; Halder et al. 2024). Common kernels include RBF, Polynomial, and Sigmoid. SVM performs well with limited training samples and is resistant to overfitting (Hassan et al. 2025). Hyperparameter tuning improves classification accuracy. The decision function of SVM is given by,

$$f(x) = \sum_{i=1}^{n} a_i y_i K(x_i, x) + b \tag{19}$$

where a_i are the Lagrange multipliers, y_i are the class labels (flooded/non-flooded), $K(x_i, x)$ is the kernel function, b is the bias term. The optimal hyperplane satisfies:

$$w^T x + b = 0 \tag{20}$$

where w is the weight vector, and x represents the feature space.

3.4. Flood inventory mapping using machine learning model

This study explores the use of historical records and remote sensing techniques in developing flood inventory maps. Data from the State Disaster Management Authority (SDMA) serves as the primary source for compiling information on past flood events. The flood inventory map is integrated with the study area map, as illustrated in Figures 6 and 7. For further analysis, the dataset is divided into two subsets: 70% for testing (68 flood points) and 30% for validation (30 flood points). Flood presence is represented by a binary classification, where 1 indicates flood sites and 0 represents non-flood sites. To minimize bias, previous studies recommend maintaining an equal number of flood and non-flood points. In line with this approach, 98 non-flood sites were selected through a random process and categorized into two groups: 70% (68 points) for testing and 30% (30 points) for validation. The dependent variable in this study includes 98 flood locations (value = 1) and 98 non-flood locations (value = 0). The machine learning models applied to these datasets were tested and validated using the 30 flood and 30 non-flood validation points, ensuring a balanced evaluation of flood susceptibility predictions.

3.5. Procedure for assessing multicollinearity in the selected flood influencing factor

Multicollinearity represents to a state or situation in regression analysis where two or more predictor variables are highly correlated, meaning that one can be linearly predicted from the others with a substantial degree of accuracy (Costache and Zaharia 2017). This condition can cause problems in the estimation of regression coefficients and make it difficult to assess the effect of individual predictors (Gorelick et al. 2017). There are several methods to detect multicollinearity, including; correlation matrix, variance inflation factor (VIF), tolerance, eigenvalues and condition index, etc. (Dano et al. 2019). Multicollinearity increases the standard errors of the regression coefficients, which can make the coefficients statistically insignificant even if they are important (Arabameri et al. 2020). The regression coefficients become highly sensitive to changes in the model. Even slight alterations in the data could result in significant changes in the calculated coefficients (Arora et al. 2021). It becomes difficult to determine the individual effect of each predictor due to their high intercorrelation. If certain predictors are highly correlated, consider removing one of them from the model (Mitra et al. 2022). Tolerance (TOL) and Variance Inflation Factor (VIF) are closely related concepts used to assess multicollinearity in multiple regression models (Ghosh et al. 2023). This study utilised Tolerance and VIF to evaluate the level of multicollinearity among the 8 identified factors that impact flooding. Tolerance and VIF, and both can be calculated as follows.

$$T_i = 1 - R_i^2. \tag{21}$$

$$VIF_i = \frac{1}{T_i}. \tag{22}$$

where 'T_i is the tolerance for the ith predictor variable. R_i^2 is the coefficient of determination when the ith variable is regressed against all the other predictor variables, and VIF_i is the VIF for the ith predictor variable'. High tolerance values (close to 1) indicate that a predictor variable is not affected by multicollinearity and has little variance explained by other predictors. Conversely, low tolerance values suggest that a variable is highly influenced by multicollinearity. A high VIF (typically greater than 10) suggests strong multicollinearity, while a low VIF indicates that a predictor variable is not highly correlated with others in the model (Debnath et al. 2023).

3.6. Model validation metrics

For flood susceptibility mapping with different machine learning models these metrics (Kappa, MAE, RMSE, RAE, Precision, Recall, F1-Measure, MCC, ROC Area, and PRC Area) assess model accuracy, reliability, and predictive performance (Debnath et al. 2024; Demissie et al. 2024; Halder et al. 2024; Prakash et al. 2024; Saleh et al. 2024; Singha et al. 2024; Tran et al. 2024). Cohen's Kappa measures the agreement between observed and predicted classifications, adjusting for chance agreement. It is given by,

$$\kappa = \frac{p_0 - p_e}{1 - p_e} \tag{23}$$

where p_0 is the observed agreement (accuracy), p_e is the expected agreement by chance. A κ value of 1 indicates perfect agreement, while 0 means no better than random guessing. RMSE measures the square root of the average squared differences between predictions and actual values. MAE measures the average magnitude of errors between predictions and actual values. RAE is the ratio of the total absolute error to the total absolute error of a simple predictor.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{24}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|(y_i - \hat{y}_i)^2\right| \tag{25}$$

$$\mathrm{RAE} = \frac{\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right|}{\sum_{i=1}^{n}\left|y_i - \overline{y}_i\right|} \tag{26}$$

where y_i is actual value. $\hat{y}_i$ is predicted value. $\overline{y}_i$ is mean of actual values. n is number of observations. Precision measures the proportion of true positives out of all predicted positives. Recall measures the proportion of true positives correctly identified. The F1-Measure is the harmonic mean of Precision and Recall. MCC is a balanced measure for binary classification, even with imbalanced classes.

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{27}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{28}$$

$$\text{F1-Measure} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{30}$$

$$\mathrm{MCC} = \frac{(\mathrm{TP} \times \mathrm{TN}) - (\mathrm{FP} \times \mathrm{FN})}{\sqrt{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}} \tag{31}$$

where TP is True Positives, TN is True Negatives, FP is False Positives, FN is False Negatives. The Receiver Operating Characteristic (ROC) Curve plots the Y = sensitivity against the X = 1-specificity at different classification thresholds. The Area Under the Curve (AUC-ROC) measures the classifier's ability to distinguish between classes. ROC-AUC range between 0 to 1. These classes are defined as follows; 1.0 (perfect classifier), 0.9–1.0 (excellent), 0.8–0.9 (good), 0.7–0.8 (fair), 0.6–0.7 (poor), 0.5–0.6 (week), < 0.5 (worse than random).

$$X = 1\text{-specificity} = 1 - \left[\frac{FP}{(FP + TN)}\right] \tag{32}$$

$$Y = \text{sensitivity} = \left[\frac{TP}{(TP + FN)}\right] \tag{33}$$

The Precision-Recall Curve (PRC) is used for imbalanced datasets, as it focuses on positive class predictions. It plots Y = Precision against X = Recall at different classification thresholds. The Area Under the PR Curve (AUC-PRC) is a key evaluation metric. PRC-AUC range between 0 to 1. These classes are defined as follows; 1.0 (perfect classifier), 0.9–1.0 (excellent), 0.8–0.9 (good), 0.7–0.8 (fair), 0.6–0.7 (poor), 0.5–0.6 (week), < 0.5 (worse than random).

$$X = \text{Recall} = \left[\frac{TP}{(TP + FN)}\right]. \tag{34}$$

$$Y = \text{Precision} = \left[\frac{TP}{(TP + FP)}\right] \tag{35}$$

4. Results

The primary thematic layers were integrated employing spatial analysis and AHP technique to generate weighted thematic layers. The following are the primary conditioning factors that are included: -

4.1. Rainfall map

Rainfall is the predominant hydrological factor commonly utilized in flood research studies. Areas that receive more rainfall than the average amount per year are more prone to flooding. Therefore, high levels of rainfall serve as an indicator of significant vulnerability to flooding (Ghosh et al. 2023). As the major area of the study falls in the Western Ghats, the average rainfall in Kerala ranges from around 1500 mm to 3000 mm, whereas our study area receives moderate to high precipitation. The rainfall distribution show that the western portion of the area receives high precipitation, while the intensity decreases as we move down the basin. This region's monthly cumulative precipitation has been catagorized into five classes: very low (866.9 to 1260.41 mm), low (1260.41 to 1601.50 mm), moderate (1601.51 to 1942.60 mm), high (1942.60 to 2277.13 mm) and very high (2277.14 to 2539.52 mm) (Figure 4(a)).

4.2. Drainage density

Drainage density is used to quantify stream spacing. Increased drainage density will dramatically increase the likelihood of flooding in a region (Halder et al. 2024). The drainage density is inversely proportional to the permeability of the land. Drainage density is influenced by various elements, including vegetation, geology, slope angle,

and soil type. The drainage density of the Periyar River basin is divided into five classes such as: very low (7.22–30.86 km/km^2), low (30.86–42.51 km/km^2), moderate (42.23–52.23 km/km^2), high (52.23–62.91 km/km^2), and very high (62.92–89.79 km/km^2) (Figure 4(b)). The majority of the region has a moderate to low drainage density.

4.3. Slope

The slope has a crucial role in hydrology, influencing surface runoff and flooding. Steeper slopes typically drain rained water faster and due to less infiltration, resulting

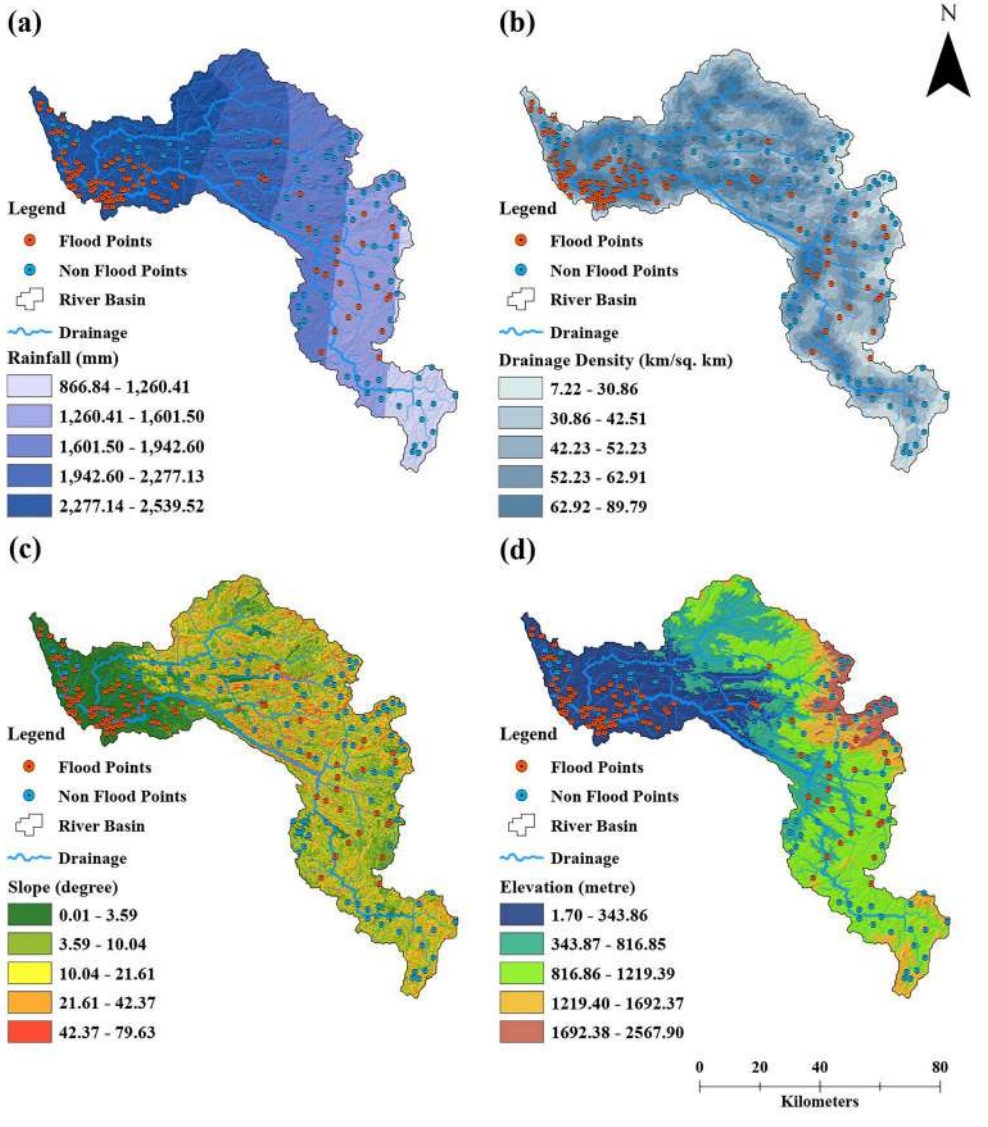


Figure 4. Flood conditioning parameters: (a) rainfall map, (b) drainage density, (c) slope, (d) elevation map of Periyar river basin, (e) TWI, (f) distance to the river, (g) soil type, (h) LULC map of Periyar river basin.

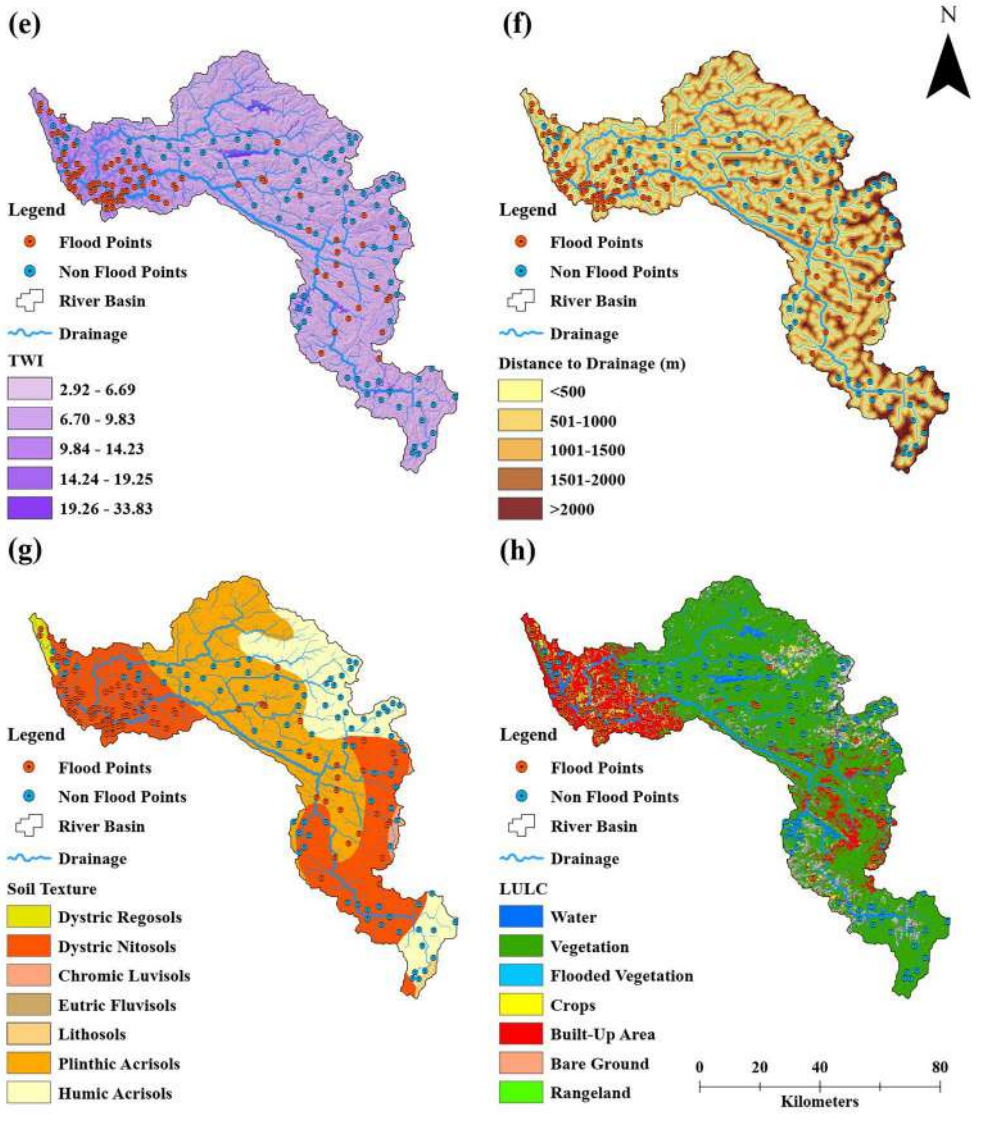


Figure 4. (Continued).

in a greater quantity of runoff (Donchyts et al. 2016). Less infiltration results in increased runoff, which enables more water to reach the lowlands. The steeper the slope, the less likely it is that flooding will occur, as there will be more surface drainage. In low-lying areas with gentler slopes, water runoff diminishes and the risk of flooding increases if there is no infiltration. The Periyar river basin is divided into five slope classes: very gentle (0.01–3.59), gentle (3.59–10.04), moderate (10.04–21.61), steep (21.61–42.37), and very steep (42.37–79.63) (Figure 4(c)). The eastern region has steeper slope values. In the western portion of the basin, namely the Ernakulam district, the slope is gentle.

4.4. Elevation

Heavy rainfall induces rapid water flow from higher-elevation places to low-lying or flat areas, which increases the chance of flooding in locations that are either flat or lower in elevation (Ke et al. 2020). The study region's elevation map is displayed in Figure 4(d). The western portion of the study area has low elevation. The elevation of the research area ranges between 0 to 2609.2 m. The elevation map was categorized into five classes, such as: 1.70–343.86 m, 343.87–816.85 m, 816.86–1219.39 m, 1219.40–1692.37 m, and 1692.38–2567.90 m.

4.5. Topographic wetness index (TWI)

The topographic wetness index (TWI) is a crucial factor in flood occurrence as it enables the spatial visualization of variations in basin moisture. As the index value increases, the amount of water accumulated also increases; conversely, as the index value decreases, the amount of accumulation decreases (Ghosh et al. 2023). The Topographic Wetness Index (TWI) of the study area (Figure 4(e)) exhibits values ranging from 2.92 to 33.83. The TWI map was classified into five distinct classes: 2.92–6.69, 6.70–9.83, 9.84–14.23, 14.24–19.25, and 19.26–33.83, representing increasing levels of potential soil moisture and surface saturation. The Topographic Wetness Index (TWI) exhibits a direct correlation with floods. Consequently, greater TWI values are assigned higher ratings, while lower TWI values are assigned lower ratings. It is evident from Figure 4(e) that the western region has a higher moisture index. The positive area comprises the wet areas, while the negative section has substantially moisture deficient.

4.6. Distance to the river

Flooding is more common in the locations that are nearest to rivers, namely in the active floodplain zones, compared to those that are farthest away from rivers. It is vital to determine the distance from the river in order to determine which surrounding areas are at risk of flooding (Mitra et al. 2022). The multiple buffer approach is used to locate distance from the river. The distance to the river parameter is categorized into five subcategories viz. $<$500 m, 501–1000 m, 1001–1500 m, 1501–2000 m, and $>$2000 m (Figure 4(f)). Typically, riverine regions are more susceptible to flooding. Therefore, areas $<$500 m will be ranked higher, as this region will be the first to experience flooding.

4.7. Soil

Soil types play a crucial role in determining the process of rainfall-runoff. The characteristics of the soil immediately restrict the movement of water into the ground, which in turn controls the process of rainfall turning into surface runoff. The soil data of the Periyar river basin is obtained from the Food and Agriculture Organization (FAO) soil map portal (https://www.fao.org/soils-portal/data-hub/soil-maps-and-databases/en/). There are three FAO soil classes reclassified and weight

have been assigned ranged from 1 to 7 based on soil texture, soil permeability, soil depth in association with its presence in hilly, moderate, or low-lying terrain (Figure 4(g)).

4.8. Land use and land cover (LULC)

The capacity of different land cover types to either absorb rainfall or aid its penetration into the soil varies. Natural vegetation, such grasslands or forests, has the ability to absorb rainfall and encourage penetration, which reduces the amount of surface runoff (Ghosh et al. 2023). On the other hand, metropolitan areas or surfaces that don't allow water to pass through easily have limited ability to absorb water. This results in more runoff on the surface and faster flow of water towards streams and rivers, which increases the chances of flooding. The Sentinel-2, 2021 satellite data was used to prepare a LULC map of the Periyar river basin. As study area lies in the Western Ghats, the majority of the basin, i.e. 64% is under vegetation, as depicted in the map (Figure 4(h)). The vegetative area is found majority in the eastern part of the basin. The majority of the highlands are located in the eastern portion of the basin, while the western portion comprises of urban/built-up areas. Built-up and rangeland cover 15% of the total area, with 3% of water bodies and crops. Built-up is mainly seen in the western part of the basin, i.e. in the low-lying regions of the basin.

4.9. Multicollinearity analysis to examine the relationship between the flood influencing factors and flood susceptibility mapping

In order to assess the presence of multicollinearity and choose variables for regression analysis, the Variance Inflation Factor (VIF) is employed. Nevertheless, VIF alone does not ascertain the variables that have an impact on the dependent variable. In order to assess the level of multicollinearity in the susceptibility and vulnerability modelling and to determine the significance of the selected flood-controlling variables, both the Variance Inflation Factor (VIF) and Tolerance (TOL) value multicollinearity measures were derived. A collinearity issue is typically detected when the Variance Inflation Factor (VIF) exceeds 10 and the Tolerance (TOL) value is below 0.1. The VIF and tolerance values for all conditioning factors in this investigation range from 10 to 0.01 (Table 4), showing that none of the 8 selected flood conditioning factors exhibit multicollinearity. The results indicate that elevation has the greatest Variance Inflation Factor (VIF) value of 2.085 and the lowest Tolerance (TOL) value of 0.480.

4.10. Flood susceptibility mapping using AHP method

The flood susceptibility map was generated using the weighted overlay method in ArcGIS platform after weights are assigned using AHP (Table 5). The flood risk map is divided into five risk categories: very low (14%), low (16%), moderate (52%), high (14%), and very high risk (4%) (Figure 5). The extremely high-risk flood zone is located in the western portion of the basin. The majority of the built-up areas are in

Table 4. Multicollinearity test (Tolerance and VIF values) of the food conditioning factors.

Collinearity statistics	Parameters	Elevation	Slope	TWI	DD	DfR	Rainfall	Soil	LULC
	Tolerance	0.48	0.822	0.978	0.732	0.793	0.77	0.798	0.969
	VIF	2.085	1.216	1.023	1.367	1.26	1.298	1.253	1.032

Table 5. Parameters used, their classification criteria and their weights using AHP.

Flood causative factors	Unit	Susceptibility class ranges	Ratings	Susceptibility class ratings	Weight (%)
TWI	Level	2.92–6.69	Very Low	1	18.7
		6.70–9.83	Low	3	
		9.84–14.23	Medium	5	
		14.24–19.25	High	7	
		19.26–33.83	Very High	9	
Elevation	m	1692.38–2567.90	Very Low	1	12.2
		1219.40–1692.37	Low	3	
		816.86–1219.39	Medium	5	
		343.87–816.85	High	7	
		1.70–343.86	Very High	9	
Slope	Degree	− 3.59	Very High	9	
		3.59–10.04	High	7	
		10.04–21.61	Medium	5	7.7
		21.61–42.37	Low	3	
		42.37–79.63	Very Low	1	
Precipitation	mm	866.84–1,260.41	Very Low	1	14.7
		1,260.41–1,601.50	Low	3	
		1,601.50–1,942.60	Medium	5	
		1,942.60–2,277.13	High	7	
		2,277.14–2,539.52	Very High	9	
Distance from River	m	<500	Very High	9	19.4
		501–1000	High	7	
		1001–1500	Medium	5	
		1501–2000	Low	3	
		>2000	Very Low	1	
LULC	Km	Water	Very High	9	3.07
		Built Area		8	
		Bare Ground		7	
		Rangeland	Medium	6	
		Flooded Vegetation		5	
		Crops		3	
		Vegetation	Very Low	1	
Drainage density	Km^2	62.92–89.79	Very High	1	21.39
		52.23–62.91	High	3	
		42.23–52.23	Medium	5	
		30.86–42.51	Low	7	
		7.22–30.86	Very Low	9	
Soil	Unitless	As layers defined (Figure 9)	Very Low- Very High	1–7	3.8

low-lying western areas; hence, the impact of the flood is particularly severe. As we approach the middle of the basin, we can observe a reduction in the severity of the risk zone, which is located in mountainous terrain. The flood risk is low to extremely low risk in the eastern portion of the basin.

Based on the flood hazard map, the western section of the basin, which is home to most of the urban population and important transportation infrastructure, is classified as having a high risk of flooding. The recent proliferation of building development in high flood risk zones has significantly reduced the infiltration of water into the earth and amplified water runoff, hence exacerbating the occurrence of floods. Based on the findings, the regions of Kothamangalam, Muvattupuzha, Kunnathunad, Aluva taluka,

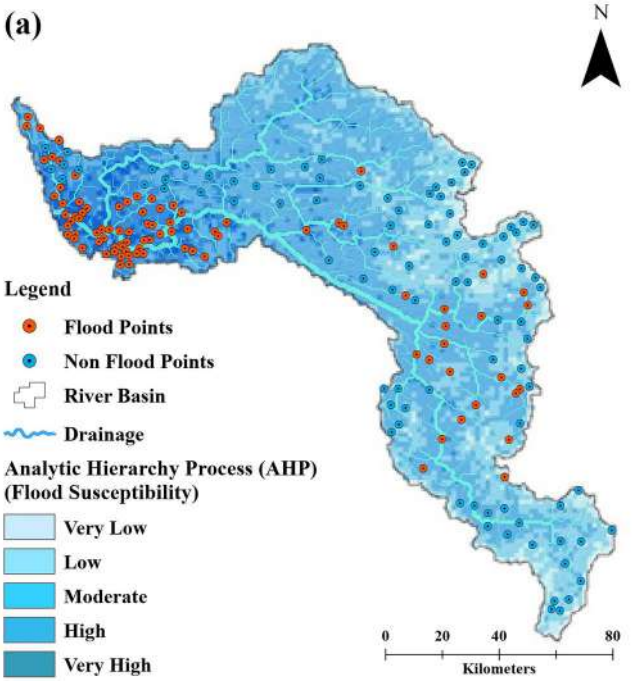


Figure 5. Flood susceptibility map of Periyar river basin using AHP method.

Perumbavoor, Karukutty, Eloor, Angamaly, Kalady, Malayattoor, Kothamangalam, Muvattupuzha, Piravam, Koothattukulam, Malayattoor, Parakadavu, Keezhmadu, Chowara, Chengamanad, Aluva, Angamaly, Nedumbassery, Mattur, Kalady Choornikkara, etc. are classified as very high to high-risk zones. The findings suggest that Aluva, Paravoor, Kodungalloor, Perumbavoor, and Kothamangalam taluk in the Ernakulam district, which are highly developed urban areas in central Kerala, are under the high to extremely high flood risk zones. Based on the past flood records of Kerala, particularly in Aluva, these areas are prone to frequent floods. The built-up area comprises 15% of the entire basin. The proportion of built-up areas within each risk zone include 15.31% under 'very high-risk zones', 42.15% as the 'high-risk zones', 31.87% as the 'moderate risk Zones', 8.65% as the 'low-risk zones' and 1.96% as 'very low-risk zones'. The majority of the built-up regions lie in the very-high to moderate risk zones (Figures 6 and 7). In order to minimize the adverse effects of flooding, it is imperative to use strategic strategies for planned building in these regions. To mitigate additional susceptibility, it is imperative to eliminate encroachments, especially those in close proximity to the river banks. By adopting these measures, the population and infrastructure in high-risk areas can be safeguarded and the potential damage from inundation can be substantially reduced, thereby ensuring their safety and protection.

4.11. Flood susceptibility mapping using machine learning approach

Flood susceptibility maps of the are developed using eight advanced machine learning models; Random Sub Space, Bagging, Radial Basis Functions, Artificial Neural

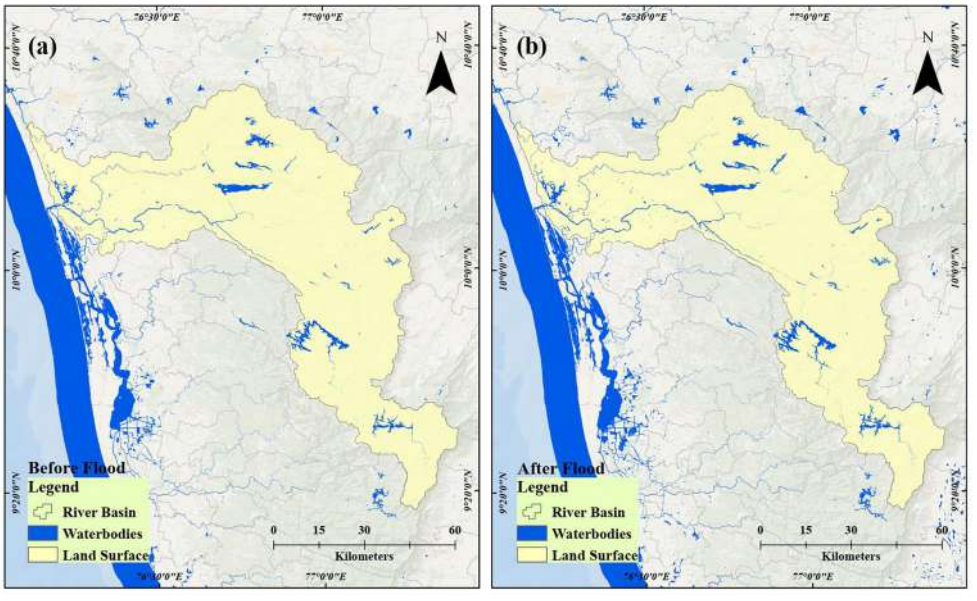


Figure 6. Pre and post flood scenario of 2018 of Periyar river basin.

Network, M5P, Additive Regression, Random Forest and Support Vector Machine. These machine learning models are implemented inside RS and GIS context. The final map is categorized into five distinct classifications, namely very low, low, moderate, high, and very high susceptibility to floods. As seen in Figures 8 and 9, the maps illustrate that the regions exhibiting significant sensitivity are mostly localized within the region and along the course of the River. Furthermore, it is evident that the categories with a significantly high susceptibility include a substantial quantity of training data points that were derived from the affected areas during the occurrence of the flood. This empirical evidence serves to validate the robust performance of the model.

For Random Sub Space shows, very low 2472.92 sq. km (38.93), low 1425.30 sq. km (22.44%), moderate 1171.09 sq. km (18.44%), high 577.73 sq. km (9.09%) and very high 705.36 sq. km (11.10%) flood susceptibility (Figure 10). For Bagging shows, very low 1415.30 sq. km (22.28%) low 1996.78 sq. km (31.43) moderate 1208.10 sq. km (19.02) high 984.57 sq. km (15.50%), and very high 747.65 sq. km (11.77%) flood susceptibility (Figure 10). For Radial Basis Functions shows, very low 1403.50 sq. km (22.09%), low 2086.97 sq. km (32.85%), moderate 1414.83 sq. km (22.27%), high 835.96 sq. km (13.16%), and very high 611.15 sq. km (9.62%) flood susceptibility (Figure 10). For Artificial Neural Network shows, very low 2204.05 sq. km (34.70%), low 1366.51 sq. km (21.51%), moderate 1008.37 sq. km (15.87%), high 991.26 sq. km (15.60%), and very high 782.22 sq. km (12.31%) flood susceptibility (Figure 10). For M5P shows, very low 1165.90 sq. km (18.35%), low 2728.58 sq. km (42.95%), moderate 1678.79 sq. km (26.43%), high 412.53 sq. km (6.49%), and very high 366.60 sq. km (5.77%) flood susceptibility (Figure 10). For Additive Regression shows, very low 813.29 sq. km (12.80%), low 2090.37 sq. km (32.91%), moderate 1697.44 sq. km (26.72%), high 1107.60 sq. km (17.44%), and very high 643.70 sq. km (10.13%) flood

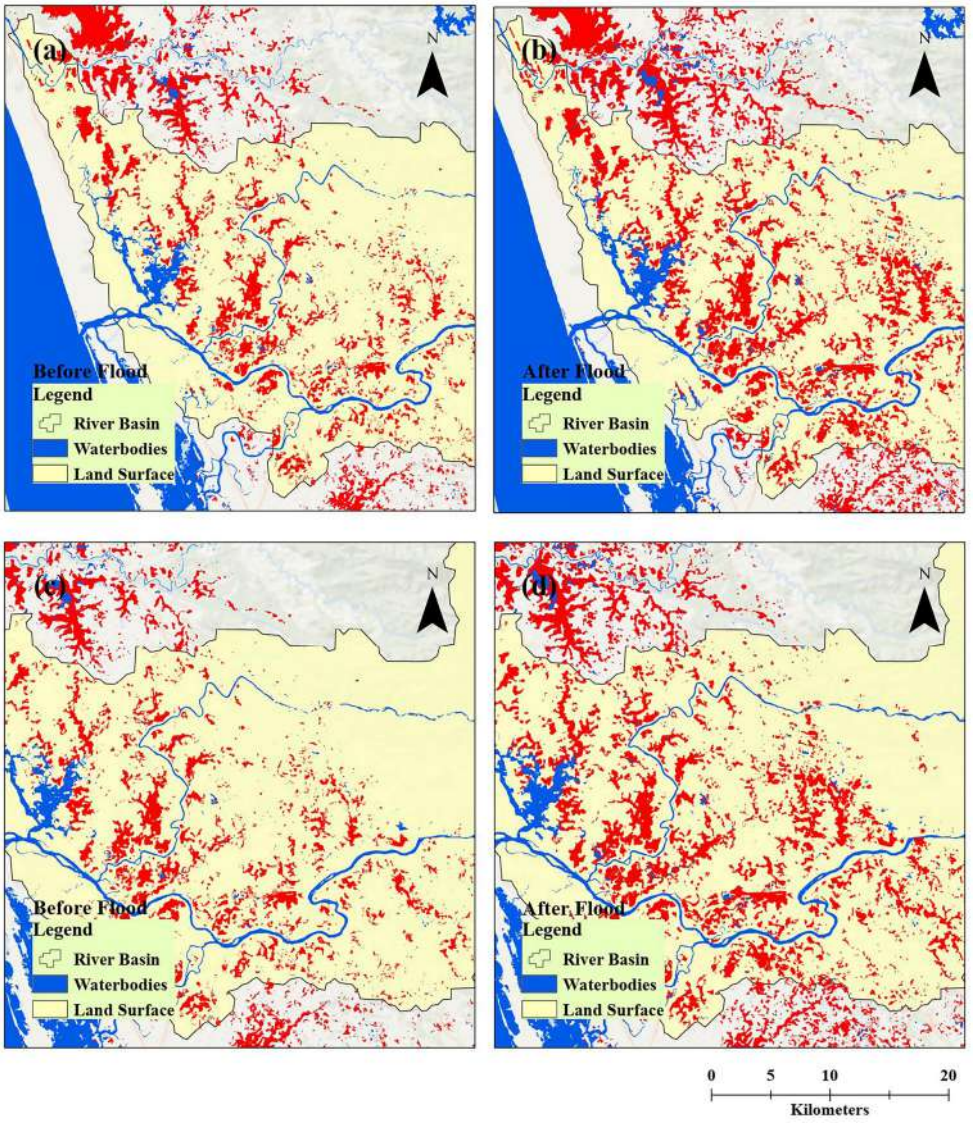


Figure 7. The built-up region under water pre and post flood scenario of 2018.

susceptibility (Figure 10). For Random Forest shows, very low 2514.55 sq. km (39.58%), low 1255.18 sq. km (19.76%), moderate 873.14 sq. km (13.75%), high 917.46 sq. km (14.44%), and very high 792.07 sq. km (12.47%) flood susceptibility (Figure 10). For Support Vector Machine shows, very low 2454.18 sq. km (38.63%), low 1267.68 sq. km (19.96%), moderate 1479.39 sq. km (23.29%), high 487.87 sq. km (7.68%), and very high 663.29 sq. km (10.44%) flood susceptibility (Figure 10) (Table 6).

4.12. Model validation

The main characteristic of any proposed model is its capacity to accurately correspond with reality and generate meaningful results (Aydin and Iban 2023). It is

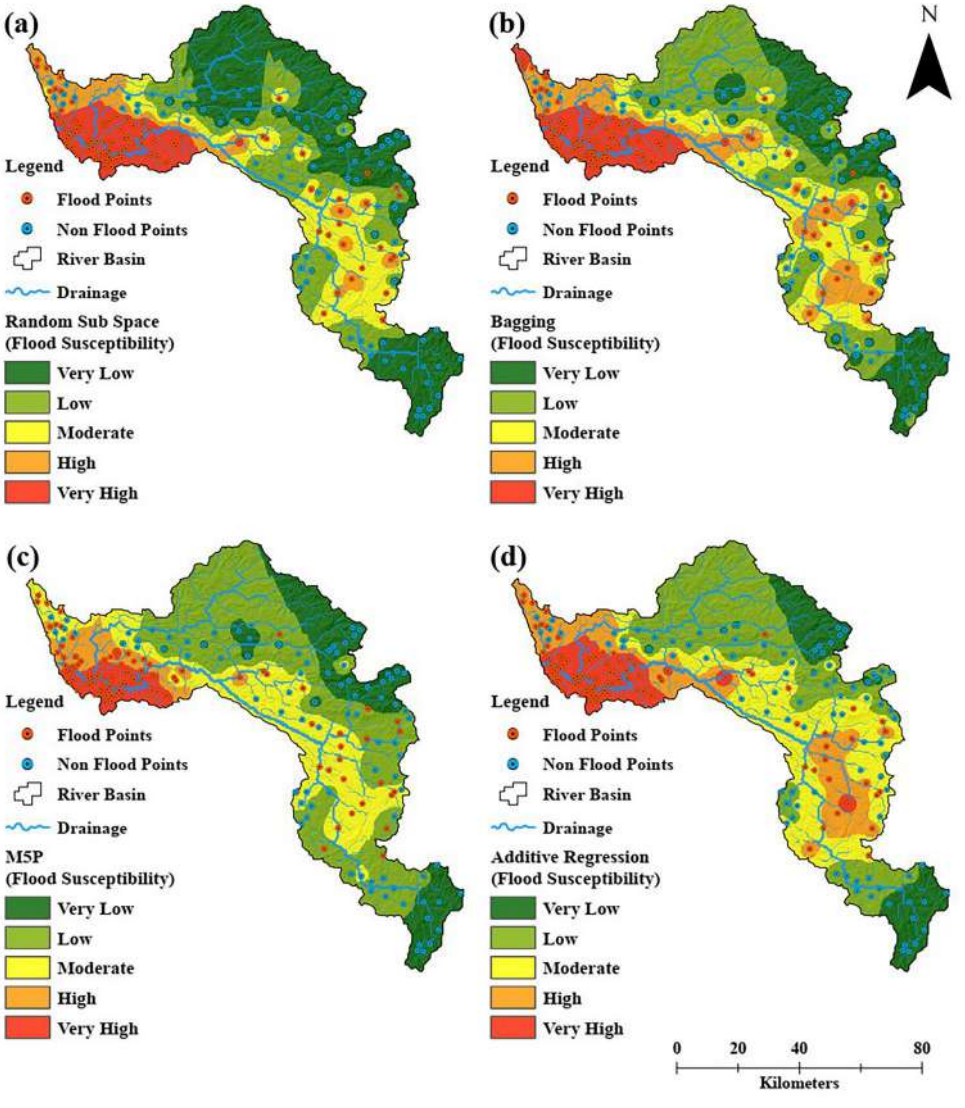


Figure 8. Flood susceptibility map. (a) Random Sub space, (b) bagging, (c) M5P, (d) additive regression.

imperative to assess the success rate of the AHP flood susceptibility mapping so that it can be validated. The compilation of the historic flood zones was based on data obtained from the Kerala state disaster management portal, as well as pre- and post-flood Sentinel-1 images. The validation approach involved the acquisition of two SAR images with VV polarization: one taken before the actual flood event (10 August 2018) and another taken after the flood event (23 August 2018) using Google Earth Engine (GEE) cloud platform. The flooded zone or sites are determined by analyzing both SAR images using the image ratio approach, with a threshold value of 1.2. By utilizing GEE to retrieve data before and after the flood, it was determined that the 2018 flood caused significant damage to the northwestern section of the basin

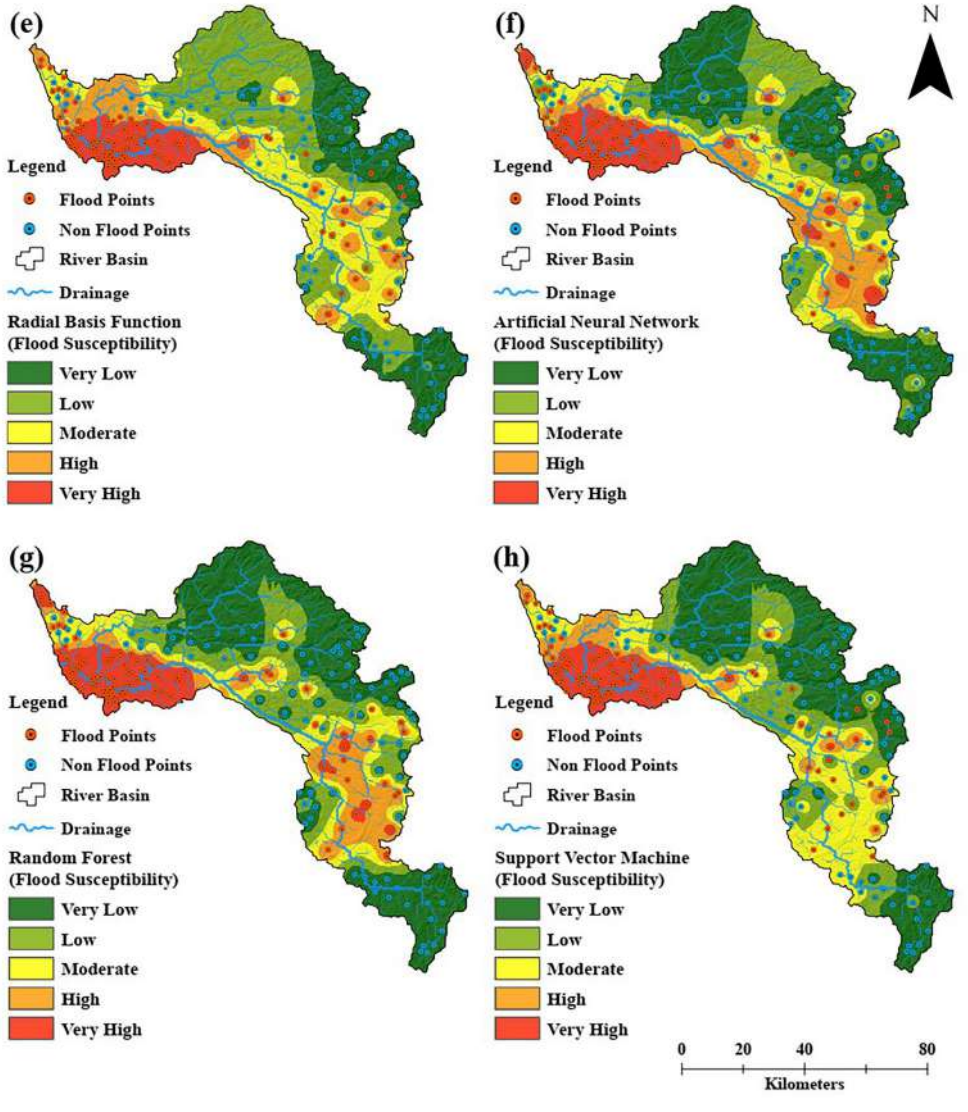


Figure 9. Flood susceptibility map. (e) Radial basis functions, (f) artificial neural network, (g) random forest, (h) Support vector machine.

(Figures 6 and 7). The acquired results are consistent with the findings of previous studies conducted by Jacinth Jennifer et al. in 2022 and Parthasarathy et al. in 2021. These results have been verified by comparing them with the maps released by the Kerala State Disaster Management Authority. Using the AUC-ROC method, this study also intends to verify the efficacy and performance of the AHP model. To validate the flood susceptibility map for the AHP model, we plotted the area under the curve - receiver operating characteristics (AUC-ROC) against the flooded sites. A total of 98 flooded spots were considered in the districts of Ernakulam, Idukki, and Thrissur with their flood water level as given by the KSDMA data. The AUC value obtained in this study is 82.5%, indicating a very accurate spatial mapping model for

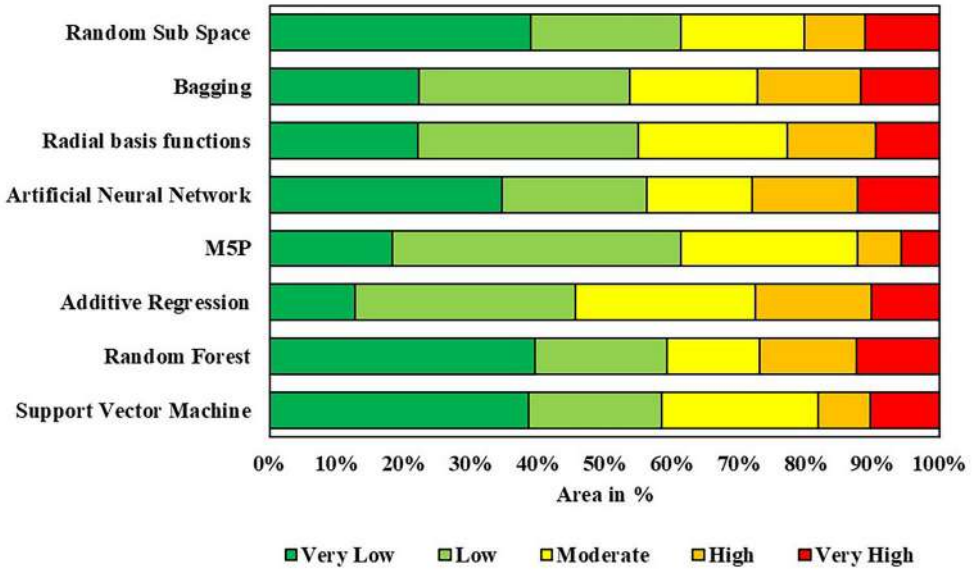


Figure 10. Spatial distribution of flood susceptibility using machine learning models.

flood susceptibility investigations (Figure 11). This indicates that the implemented flood susceptibility method is highly effective in precisely predicting the probability of a region being inundated.

Based on the validation metrics (Kappa, MAE, RMSE, RAE, Precision, Recall, F1-Measure, MCC, ROC Area, and PRC Area), a comprehensive analysis of the eight machine learning models have been shown in Table 7. For Random Sub Space model, Kappa (0.780) shows strong agreement with actual labels. MAE (0.214), RMSE (0.274), RAE (0.429) shows moderate error rates. Precision (0.920), Recall (0.860), F1-Measure (0.889), shows high precision and recall with balanced performance. MCC (0.782) shows strong correlation between predicted and actual labels. For Bagging model, Kappa (0.857) shows very strong agreement. MAE (0.200), RMSE (0.259), RAE (0.400) shows slightly better than Random Sub Space. Precision (0.926), Recall (0.935), F1-Measure (0.930) shows excellent detection ability. MCC (0.857) shows strong and reliable predictions. For Radial Basis Functions model, Kappa (0.737) shows moderate agreement. MAE (0.232), RMSE (0.315), RAE (0.465) shows higher errors than previous models. Precision (0.906), Recall (0.828), F1-Measure (0.865) shows good, but recall is slightly lower. MCC (0.740) shows decent correlation, but weaker than Bagging. For Artificial Neural Network model, Kappa (0.857) shows excellent agreement. MAE (0.191), RMSE (0.305), RAE (0.382) shows lower MAE but higher RMSE than Bagging. Precision (0.976), Recall (0.882), F1-Measure (0.927) shows high precision but slightly lower recall. MCC (0.862) shows best performance among models. For M5P model, Kappa (0.714) shows moderate agreement. MAE (0.231), RMSE (0.314), RAE (0.462) shows similar performance Radial Basis Functions. Precision (0.860), Recall (0.860), F1-Measure (0.860) shows balanced, but lower than ANN. MCC (0.714) shows moderate predictive power. For Additive Regression model, Kappa (0.769) shows strong agreement between predicted and

Table 6. Spatial distribution of flood susceptibility models.

Model	Random sub space		Bagging		Radial basis functions		Artificial neural network		M5P		Additive regression		Random forest		Support vector machine	
Susceptibility Level	Area in sq. km	Area in %	Area in sq. km	Area in %	Area in sq. km	Area in %	Area in sq. km	Area in %	Area in sq. km	Area in %	Area in sq. km	Area in %	Area in sq. km	Area in %	Area in sq. km	Area in %
Very Low	2472.92	38.93	1415.30	22.28	1403.50	22.09	2204.05	34.70	1165.90	18.35	813.29	12.80	2514.55	39.58	2454.18	38.63
Low	1425.30	22.44	1996.78	31.43	2086.97	32.85	1366.51	21.51	2728.58	42.95	2090.37	32.91	1255.18	19.76	1267.68	19.96
Moderate	1171.09	18.44	1208.10	19.02	1414.83	22.27	1008.37	15.87	1678.79	26.43	1697.44	26.72	873.14	13.75	1479.39	23.29
High	577.73	9.09	984.57	15.50	835.96	13.16	991.26	15.60	412.53	6.49	1107.60	17.44	917.46	14.44	487.87	7.68
Very High	705.36	11.10	747.65	11.77	611.15	9.62	782.22	12.31	366.60	5.77	643.70	10.13	792.07	12.47	663.29	10.44
Area	6352.40	100.00	6352.40	100.00	6352.40	100.00	6352.40	100.00	6352.40	100.00	6352.40	100.00	6352.40	100.00	6352.40	100.00

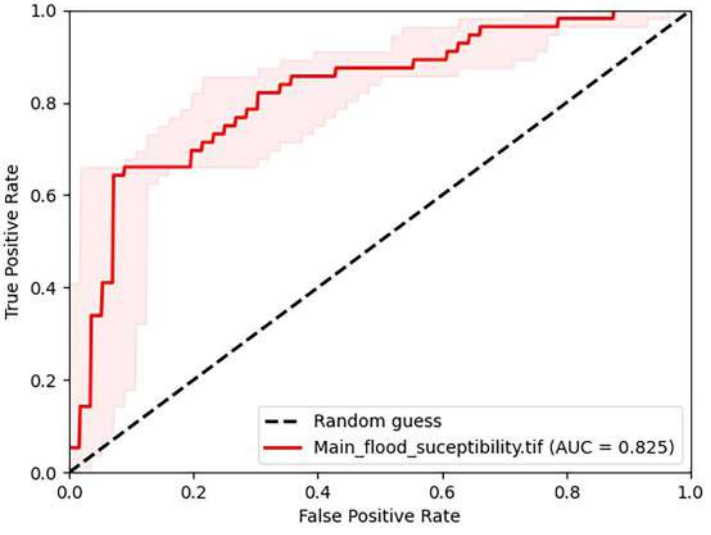


Figure 11. ROC curve for validation of predicted map using AHP method.

Table 7. Evaluation of flood susceptibility models using validation dataset.

Model	Kappa	MAE	RMSE	RAE	Precision	Recall	F1-measure	MCC	ROC area	PRC area
Random sub space	0.780	0.214	0.274	0.429	0.920	0.860	0.889	0.782	0.974	0.979
Bagging	0.857	0.200	0.259	0.400	0.926	0.935	0.930	0.857	0.985	0.987
Radial basis functions	0.737	0.232	0.315	0.465	0.906	0.828	0.865	0.740	0.937	0.953
Artificial neural network	0.857	0.191	0.305	0.382	0.976	0.882	0.927	0.862	0.953	0.967
M5P	0.714	0.231	0.314	0.462	0.860	0.860	0.860	0.714	0.946	0.953
Additive REGRESSION	0.769	0.226	0.299	0.452	0.900	0.871	0.885	0.770	0.960	0.964
Random Forest	0.682	0.183	0.298	0.366	0.864	0.817	0.840	0.683	0.932	0.950
Support vector machine	0.548	0.278	0.390	0.556	0.745	0.849	0.794	0.554	0.882	0.886

actual values. MAE (0.226), RMSE (0.299), RAE (0.452) shows moderate error rates. Precision (0.900), Recall (0.871), F1-Measure (0.885) shows good balance of precision and recall. MCC (0.770) shows strong correlation, indicating reliable predictions. For Random Forest model, Kappa (0.682) shows Moderate agreement. MAE (0.183), RMSE (0.298), RAE (0.366) shows lowest MAE but RMSE is like Additive Regression. Precision (0.864), Recall (0.817), F1-Measure (0.840) shows slightly weaker recall than Additive Regression. MCC (0.683) shows moderate correlation. For Support Vector Machine model, Kappa (0.548) shows Weakest agreement among the models. MAE (0.278), RMSE (0.390), RAE (0.556) shows highest errors, indicating weaker performance. Precision (0.745), Recall (0.849), F1-Measure (0.794) shows good recall, but precision is lower. MCC (0.554) shows weakest correlation among the models.

When validating a machine learning model, ROC-AUC and PRC-AUC are used to measure classification performance (Figures 12 and 13). For Random Sub Space model, ROC Area (0.974), PRC Area (0.979) shows excellent classification capability. For Bagging model, ROC Area (0.985), PRC Area (0.987) shows almost perfect classification. For Radial Basis Functions model, ROC Area (0.937), PRC Area (0.953) shows slightly lower than previous models. For Artificial Neural Network model, ROC Area (0.953), PRC Area (0.967) shows strong performance. For M5P model,

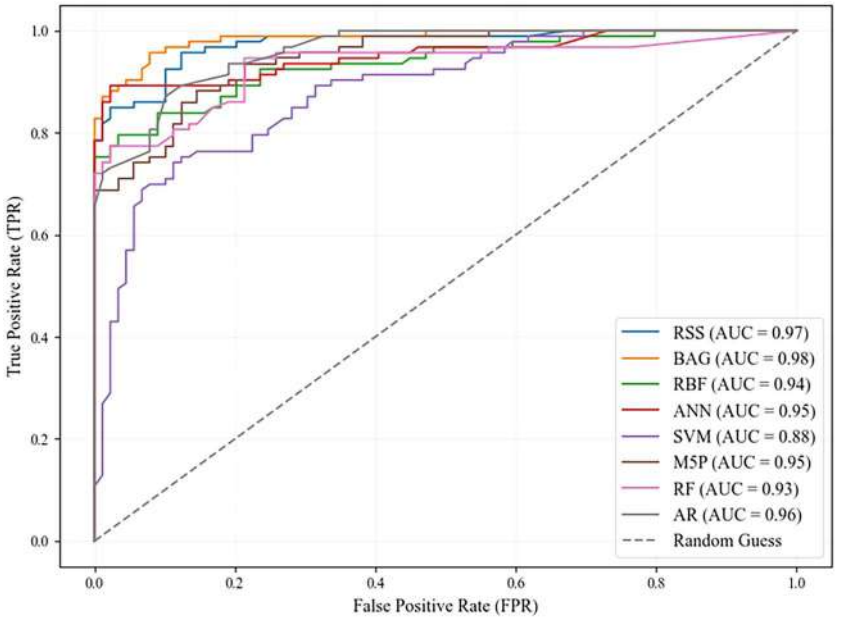


Figure 12. Validation of the of flood susceptibility machine learning models using ROC curves.

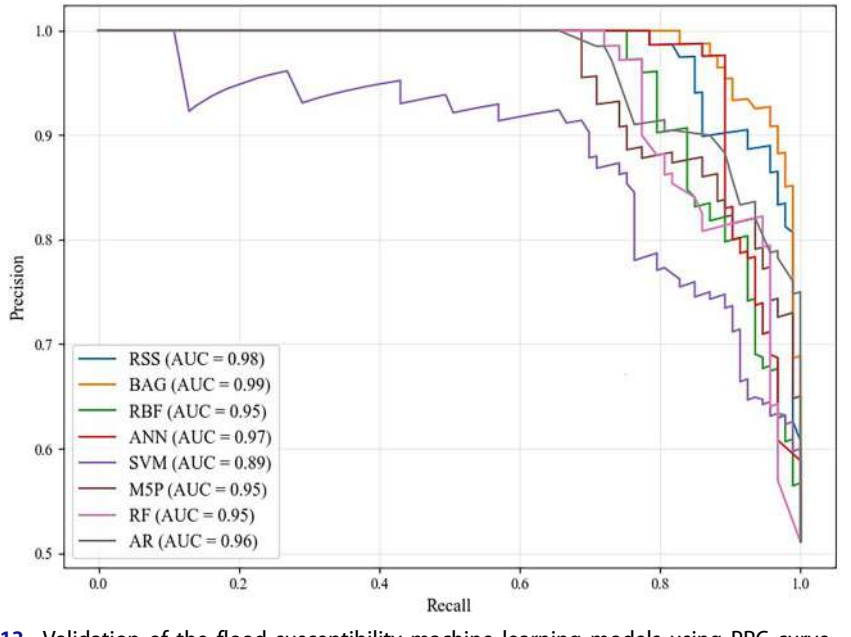


Figure 13. Validation of the flood susceptibility machine learning models using PRC curve.

ROC Area (0.946), PRC Area (0.953) shows good classification performance. For Additive Regression model, ROC Area (0.960), PRC Area (0.964) shows excellent classification performance. For Random Forest model, ROC Area (0.932), PRC Area (0.950) shows good classification capability. For Support Vector Machine model, ROC Area (0.882), PRC Area (0.886) shows decent but lower than other models.

In this study, we found that Bagging is the best overall model due to high reliability, accuracy, and minimal errors. Artificial Neural Network is the second-best model, slightly lower recall but outstanding precision. Random Sub Space good balance between precision and recall, but slightly weaker than Artificial Neural Network. Additive Regression well-performing model, but not as strong as Bagging or Artificial Neural Network. Radial Basis Functions performs well but has higher errors and weaker recall. Random Forest performs well but has lower recall and MCC than Additive Regression. M5P performs moderately well but weaker than Additive Regression. Support Vector Machine is the weakest model for this dataset, due to higher errors and lower precision.

5. Discussion

The flood susceptibility analysis of the Periyar River basin was conducted using a knowledge-based MCDA-AHP technique combined with multiple Machine Learning models, integrated within RS and GIS environment. This approach assessed flood susceptibility by considering key flood conditioning factors that contribute to flood generation, including, elevation, slope, TWI, drainage density, rainfall, distance from the river, soil type, and LULC. The study aligns with SDG 6 (Clean Water and Sanitation) by assessing flood risks that can impact water quality and sanitation infrastructure, helping to protect freshwater sources from contamination. Various works have been done previously in the Periyar River basin to estimate its river water discharge, flood susceptibility and flood risk mapping. The previous works are; Kumar and Jayarajan (2021) utilize SAR data and Landsat imagery, combined with GIS, can be effectively used to map, monitor, and assess floodwater distribution in the flood-prone regions of the lower Periyar River basin. Kumar et al. (2023) developed hybrid model by coupling SWAT (1D river flow models) with iRIC (2D floodplain flow models) to show flood conditions and to evaluate the flood-prone areas. Aneesh and Thomas (2024) utilizing advanced modelling techniques, particularly HEC-RAS for estimating region's susceptibility to devastating floods. Sreelakshmi et al. (2021) has selected 10 flood conditioning parameters and applied Weighted Overlay Analysis for estimating flood risk assessment. Geetha and Madhu (2024) has selected 10 flood conditioning parameters and applied different machine learning algorithms for flood susceptibility model; LR (Logistic Regression), SVM (Support Vector Machine), NB (Naïve Bayes), RF (Random Forest), AdaBoost (Adaptive Boosting), GB (Gradient Boosting), XGBoost (Extreme Gradient Boosting) to evaluate flood susceptibility in Periyar River basin. AUC-ROC Curve values are LR (0.90), SVM (0.88), NB (0.90), RF (0.94), AdaBoost (0.87), GB (0.92), and XGBoost (0.92). In this work knowledge-based MCDA-AHP and Machine Learning Models have been used to estimate flood susceptibility mapping. AUC-ROC Curve values of various models are AHP (0.825), Random Sub Space (0.974), Bagging (0.985), Radial Basis Functions (0.937), Artificial Neural Network (0.953), M5P (0.946), Additive Regression (0.960), Random Forest (0.932), and Support Vector Machine (0.882) shows much higher accuracy than that of previous work. Various model validation metrices (Kappa, MAE, RMSE, RAE, Precision, Recall, F1-Measure, MCC, and PRC Area) have been used to investigate

model accuracy. PRC curve has been shown which add novelty of the work. Gain Ratio Attribute (feature importance) of the machine learning models shows that these parameters have influenced most in obtaining the flood susceptibility mapping; LULC (0.235), Soil (0.158), Rainfall (0.134), Drainage density (0.134), TWI (0.131), Elevation (0.127), Slope (0.125), and Distance to drainage (0.101).

The model validation results of these integrated RS and GIS combined AHP and ML methodologies differ from the other previous similar methodological studies, achieve significant accuracy for flood susceptibility prediction (AHP, ROC-AUC > 0.825), like, Mitra et al. (2022) (AHP, ROC-AUC = 0.728), Ghosh et al. (2023) (AHP, ROC-AUC = 0.749), Saikia et al. (2024) (AHP, ROC-AUC = 0.697), Varma et al. (2025) (AHP, ROC-AUC = 0.668). It's also achieved significant accuracy for flood susceptibility prediction model validation results from the other previous different methodological studies (Machine Learning Algorithms, ROC-AUC > 0.950), like, Janizadeh et al. (2021) (Bayesian Additive Regression Tree, ROC-AUC = 0.860), Al-Areeq et al. (2023) (Light Gradient Boosting, ROC-AUC = 0.945), Abba et al. (2024) (Long Short-Term Memory, ROC-AUC = 0.863), Al-Kindi and Alabri (2024) (CatBoost, ROC-AUC = 0.910), (Support Vector Machine, ROC-AUC = 0.854), Prince et al. (2024) (Maximum Entropy Model, ROC-AUC = 0.910), Singha et al. (2024) (Random Forest, ROC-AUC = 0.847), Ahmad et al. (2025) (Extreme Gradient Boosting, ROC-AUC = 0.893), Kundu and Mondal (2025) (Logistic Regression, ROC-AUC = 0.894).

The integration of Remote Sensing (RS), Geographic Information Systems (GIS), Multi-Criteria Decision Analysis using the Analytic Hierarchy Process (MCDA-AHP), and advanced Machine Learning (ML) techniques presents an innovative and robust methodology for flood susceptibility zonation. This comprehensive approach aligns strongly with SDG 11 (Sustainable Cities and Communities) by offering data-driven insights for urban planning, infrastructure development, and disaster resilience. By combining expert-based decision-making frameworks with automated predictive algorithms, the model enhances the accuracy and adaptability of flood risk assessment in complex environments such as the Periyar River Basin. The study further supports SDG 13 (Climate Action) by improving climate resilience and disaster preparedness through scientifically grounded modelling and real-time analytics. It empowers policymakers and planners with spatially explicit flood risk information, enabling the formulation of proactive strategies, including zoning regulations, land-use planning, early warning systems, and resource allocation mechanisms. These strategies are vital for minimizing the socio-economic impacts of flood events and enhancing emergency preparedness. In addition, the methodology contributes to SDG 15 (Life on Land) by facilitating sustainable land management, reducing soil erosion, and protecting terrestrial ecosystems from flood-induced degradation. It also aligns with SDG 6 (Clean Water and Sanitation) by supporting flood-resilient water infrastructure and protecting water quality through informed watershed and drainage planning. Moreover, the integration of geospatial technologies and innovative computational models directly addresses SDG 9 (Industry, Innovation, and Infrastructure) by promoting the use of advanced, resilient infrastructure planning tools and fostering technological innovation in environmental risk assessment. The approach also indirectly supports SDG

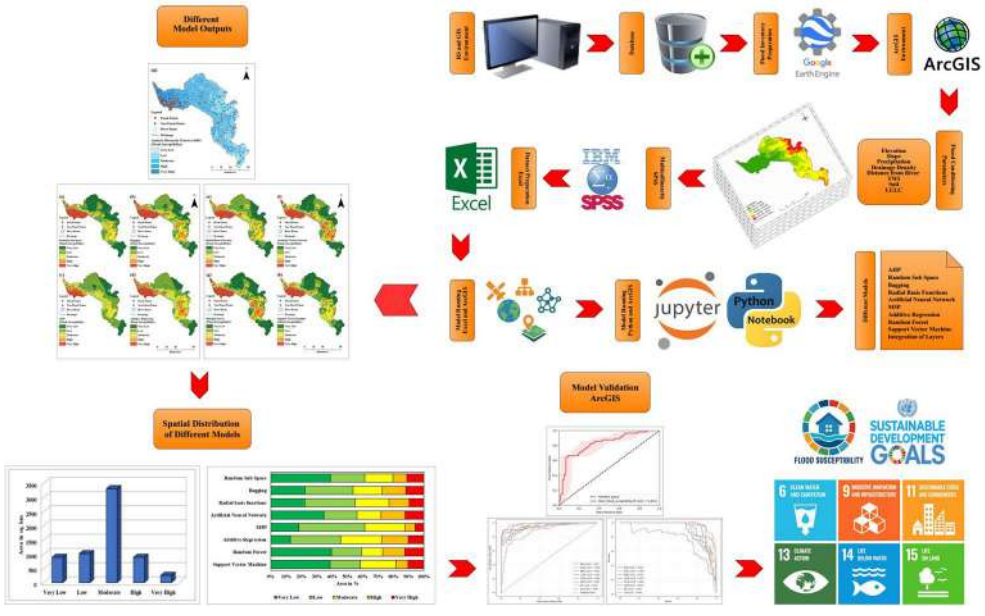


Figure 14. Conceptual framework illustrating the linkage between flood susceptibility mapping and relevant sustainable development goals (SDGs).

14 (Life Below Water) by mitigating flood-related runoff and pollution risks to downstream aquatic ecosystems and wetlands. Overall, this research provides a scientifically rigorous and practically applicable flood risk assessment framework that advances sustainable development, disaster risk reduction, and environmental conservation objectives in vulnerable and ecologically sensitive regions (Figure 14).

The integration of the Analytical Hierarchy Process (AHP) and Machine Learning (ML) techniques offers a complementary approach for flood susceptibility mapping, combining expert knowledge with data-driven prediction. AHP is advantageous due to its structured decision-making framework, which allows for the incorporation of both quantitative and qualitative criteria based on expert judgment. It is particularly useful in data-scarce environments and ensures transparency and consistency in assigning weights to multiple flood conditioning factors. However, AHP is subjective in nature, relying heavily on expert opinions, which can introduce bias and reduce repeatability. Additionally, the method may oversimplify complex hydrological interactions due to its linear hierarchical structure. On the other hand, Machine Learning algorithms offer high accuracy, automation, and adaptability, capable of capturing non-linear relationships among multiple variables through training on large datasets. ML models such as Random Forest, Support Vector Machine, and Artificial Neural Networks can handle high-dimensional data efficiently and generate predictive susceptibility maps with minimal human intervention. Nevertheless, their effectiveness is contingent upon data availability and quality, and they often lack interpretability, making it challenging to understand the rationale behind model decisions. By integrating AHP with ML techniques, this study harnesses the strengths of both approaches—AHP for expert-informed factor prioritization and ML for robust spatial

prediction—to produce a reliable and comprehensive flood susceptibility mapping framework for the Periyar River Basin.

The flood susceptibility maps generated in this study have significant practical value for disaster risk management, urban planning, and policy formulation. Local authorities and disaster management agencies can use these maps as strategic planning tools to identify high-risk zones and prioritize them for targeted flood mitigation interventions. For instance, areas categorized as highly susceptible—such as parts of Aluva, Paravoor, Kodungalloor, Perumbavoor, and Kothamangalam—can be earmarked for the construction of flood protection infrastructure, including embankments, retention basins, and improved drainage systems. These maps also support risk-sensitive land-use planning by guiding zoning regulations that restrict construction in flood-prone areas, thus reducing future exposure. Moreover, emergency response planning can be significantly enhanced, as the maps allow decision-makers to identify suitable locations for flood shelters, relief distribution centres, and evacuation routes in advance. In rural and agricultural zones, the susceptibility analysis can help in crop insurance planning and seasonal advisory services, reducing economic losses for farmers. Additionally, public awareness programs can be more effectively targeted to vulnerable communities identified through the mapping. Ultimately, these spatially explicit flood maps provide the basis for an integrated flood risk management strategy that supports climate resilience, sustainable development, and alignment with national and international disaster risk reduction frameworks.

6. Conclusion

In this study, knowledge-driven Analytical Hierarchy Process (AHP) and data-driven Machine Learning (ML) techniques were integrated with Remote Sensing (RS) environment and Geographic Information System (GIS) tools to develop a comprehensive flood susceptibility mapping framework for the Periyar River Basin. Eight key flood conditioning parameters—rainfall, elevation, slope, drainage density, distance from the river, topographic wetness index (TWI), soil type, and land use/land cover (LULC)—were analyzed to identify areas at varying degrees of flood vulnerability. The resulting flood susceptibility map revealed that approximately 971.64 sq. km (18% of the basin), primarily located in the western part of the study area, is highly prone to flooding. These high-risk zones are characterized by steep terrain, dense drainage networks, and the confluence of several tributaries into the main river. Low-lying and densely populated regions, especially in the talukas of Aluva, Paravoor, Kodungalloor, Perumbavoor, and Kothamangalam in Ernakulam district, as well as Irrinjalakuda in Thrissur district, are most severely affected.

The predictive accuracy of the models was evaluated using the Area Under the Curve–Receiver Operating Characteristic (AUC-ROC) method. The results confirmed the robustness of the approach, with AUC values of 0.825 for AHP, and high predictive accuracies for all ML models; Random Sub Space (0.974), Bagging (0.985), Radial Basis Function (0.937), Artificial Neural Network (0.953), M5P (0.946), Additive Regression (0.960), Random Forest (0.932), and Support Vector Machine (0.882).

These findings validate the reliability of the integrated methodology in accurately delineating flood-prone zones.

To reduce flood risk and enhance resilience, the study recommends a combination of structural and non-structural mitigation measures in high-risk zones. These include restricting unregulated urban development, improving access to flood shelters and emergency medical services, implementing sustainable floodplain management practices, promoting flood insurance schemes, and launching targeted public awareness programs. In urban areas, building approvals in flood-prone zones should be subject to strict flood protection criteria. Conversely, low-risk areas identified through susceptibility mapping can serve as strategic locations for flood relief camps and emergency response centres.

This research strongly supports and contributes to the achievement of several United Nations Sustainable Development Goals (SDGs). It aligns with SDG 6 (Clean Water and Sanitation) by promoting improved flood-resilient water infrastructure and watershed management. It addresses SDG 9 (Industry, Innovation, and Infrastructure) through the application of advanced geospatial technologies and predictive modelling for resilient infrastructure planning. The findings directly contribute to SDG 11 (Sustainable Cities and Communities) by supporting evidence-based urban planning, disaster preparedness, and risk-sensitive land-use policies. Through climate-informed modelling, the study also reinforces SDG 13 (Climate Action) by enhancing adaptive capacity and disaster risk reduction. By identifying zones that threaten downstream ecosystems through runoff and pollution, it indirectly supports SDG 14 (Life Below Water). Finally, the study advances SDG 15 (Life on Land) by advocating sustainable land use, preventing soil erosion, and protecting natural habitats from flood-induced degradation.

The flood susceptibility maps developed in this research are critical tools for disaster preparedness, emergency response planning, and sustainable development. They provide actionable insights for local governments, urban planners, and disaster management authorities, enabling informed decision-making and resource allocation. By integrating expert knowledge and machine learning capabilities, this study offers a replicable and scalable framework for flood risk assessment in other vulnerable river basins across India and beyond, significantly contributing to future climate resilience and sustainable hazard management.

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Author contributions

PGN, RSM: Conceptualization, Data curation, Methodology, Visualization, Writing – original draft. DS: Visualization, Investigation, Writing – review and editing. UC: Data curation, Formal analysis. SD: Supervision, Methodology, Visualization, Validation, Project administration. TPS: Writing – review and editing. AG: Methodology, Investigation, Validation, Writing – original draft.

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Data availability statement

The datasets used and examined in this study can be obtained from the corresponding author upon request.

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